Auditable Search for Agentic Quant Research

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Abstract

Frontier LLM agents can search large hypothesis spaces cheaply, and in mathematics that search yields results because a formal checker rejects every candidate that does not follow from the stated definitions. This paper proposes a design for retail quantitative research that borrows the structure but not the certainty: agents generate trading hypotheses freely, and a small controlled kernel enforces specific program-level invariants and keeps a complete, auditable record of what every agent saw. A typed strategy language and a controlled evaluator enforce constraints on data availability, order timing, and unit consistency; an experiment ledger records every evaluation and every result disclosed to an agent; and the whole adaptive search procedure is tested on simulated markets with no edge before any real result is trusted.

We argue that the binding constraint on honest search is the number of independent observations, not ideas or compute, and set the scope accordingly. The design has two research tracks sharing one ledger mechanism. A backtestable track searches for systematic strategies across many markets, including the stocks, indices, and commodities now listed as builder-deployed (HIP-3) markets on Hyperliquid, using the long histories of the underlying assets rather than the venue's short one. A forward-only track has agents register event-driven macro theses before their outcomes are known, because an LLM's judgment can be evaluated reliably only on periods after its training cutoff. In both tracks, inference uses only data that played no part in selection: search-period statistics choose candidates, and an untouched later period judges them. Data sources are typed declarations that agents can write strategies against before any feed exists; a feed binds to a declaration only after certification. A reference kernel in Lean 4 proves the language's core invariants and is differentially tested against the production engine. Formal methods can make certain errors impossible; they cannot turn noisy historical evidence into a correctness certificate, so whether the system controls false discoveries remains a claim to be demonstrated, not assumed.

1 Motivation

Agentic mathematics works as a search problem with a learned heuristic. Many subagents propose candidate lemmas, proof steps, or constructions at high throughput. The LLM acts like the heuristic in classical tree search (GOFAI-style best-first search, or MCTS in AlphaGo-like systems): it decides which branches look promising, not which are correct.

Correctness comes from somewhere else. A proof assistant such as Lean has a small trusted kernel that checks whether each candidate proof follows from the formal definitions, cheaply and mechanically. Because wrong guesses cost almost nothing, the heuristic is allowed to be wild. The kernel's guarantee is also narrower than it looks: it certifies that a proof matches the formal statement, not that the formal statement captures the intended question. A trading evaluator has the same weakness in a stronger form, since it can compute a perfectly correct result for an incorrectly modeled market.

The question for retail traders is how much of this structure carries over to strategy research, with agents as tireless research assistants and formal constraints as part of the kernel. Retail cannot match institutional data, latency, or headcount, but agents lower the cost of research labor. Many current agentic trading tools focus on order entry through a chat interface; this paper is about automating and auditing the research loop instead. Whether agents generate better hypotheses than simpler search methods is itself an open empirical question that the roadmap tests directly.

2 The core obstacle: a noisy, exploitable verifier

The analogy to mathematics breaks at the verifier. A proof checker is exact and cannot be fooled. A backtest is a noisy estimate on non-stationary data, and searching harder against it produces better-looking results whether or not any edge exists.

A pass/fail filter does not avoid this. Accepting only strategies above a Sharpe threshold is still selection. In an illustrative Gaussian example, searching a thousand null strategies and ten modest real edges accepts mostly noise; Appendix A.1 derives the example and the noise ceiling below. Agents make this worse in three ways:

1. **Hidden trials.** An agent that tweaks a parameter, looks at the result, and tweaks again has run many tests while submitting one.
2. **Unclear effective N.** A thousand near-duplicates are neither 1,000 independent tests nor one.
3. **Non-stationarity.** Even a correctly validated edge only shows the edge existed in the sample period.

The design goal follows: the system must make the complete history of what agents saw and decided auditable, so that the selection procedure can be evaluated as a whole rather than one test at a time.

2.1 The search budget

The approximate noise ceiling $h(N, T)$ (Appendix A.1) rises with trials and falls with years of data. Table 1 evaluates that derived illustration, not a decision rule; the gate is in Section 8.

Effective trials N	3 years	10 years	20 years
10	1.24	0.68	0.48
1,000	2.15	1.18	0.83
100,000	2.77	1.52	1.07

Table 1: Approximate noise ceiling $h(N, T)$ in annualized Sharpe units.

For the chosen illustrative edge range in Appendix A.1, short histories leave little search budget. Breadth is the second budget. Grinold's fundamental law [13] says that the information ratio is roughly skill per independent bet times the square root of independent bets per period. A dozen co-moving crypto assets offer few independent bets; a multi-asset universe may offer more, but breadth must be measured from strategy-return covariance, not counted as instruments.

Time lowers the noise ceiling, breadth raises the signal of a real edge, and trials raise the ceiling. A search is affordable only where the first two are large. This is why the scope of this paper extends

beyond the crypto pilot to many asset classes with long histories (Section 11), and why open-ended exploration requires a split-sample design that spends its multiplicity budget only at confirmation (Section 10).

3 Architecture: a small trusted kernel

The system splits into an untrusted generator and a trusted kernel, and the only way to obtain a backtest result is through the kernel (Figure 1). The forward-only thesis track (Section 12) uses the same append-only ledger mechanism, in its own database, but has no backtest path at all.

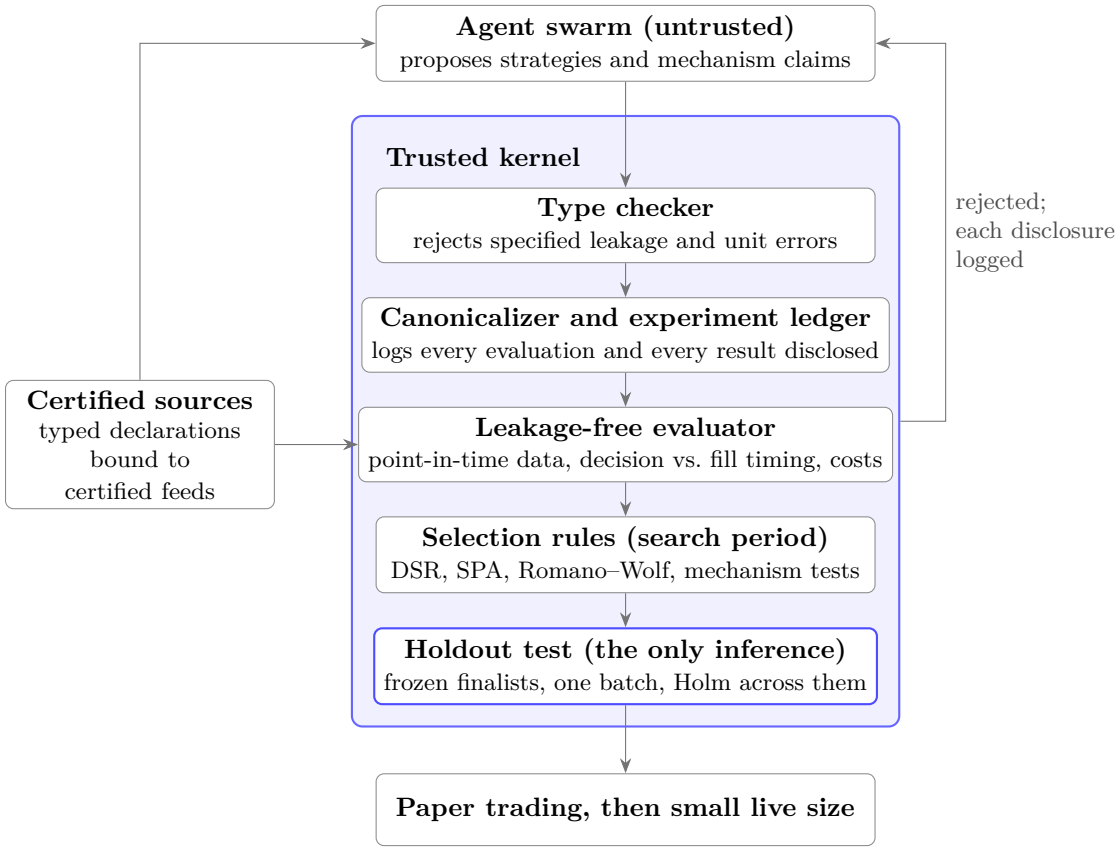


Figure 1: Agents propose freely; results come only through the kernel.

Agents propose strategies with mechanism claims; the kernel type-checks, canonicalizes and logs, evaluates, and gates them. Every result shown to an agent, including for rejected candidates, is recorded in the ledger, because it can shape what the agent tries next.

The kernel's source code should be small, but the trusted computing base is larger than the kernel. Its correctness also depends on:

- historical data reconstruction and ingestion, including revisions and timestamps;
- source certificates and the tests that produce them (Section 5);
- the DSL compiler and runtime, and any proof checker used to verify them (Section 6);
- the market simulator, fee model, and execution model;

- the statistical implementations and the ledger's storage.

Each of these needs stated invariants and tests. "Results come only through the kernel" is a deployment requirement, not a property of the design: agents must run in an environment with no direct access to market data or arbitrary compute over it. An agent that can load a CSV and run its own backtest can do unlogged research, and nothing in the kernel would know.

The deployment therefore needs a stated threat model: the channels through which an agent could learn about data or results, such as files, network access, tools, error messages, caches, other agents, and earlier sessions, and the control that closes each. The pilot runs research agents in an operating-system sandbox whose only tool is a client for a daemon that fixes the data store, enforces the evaluation budget, and discloses a fixed set of fields. Tool permissions alone do not close these channels, so the operating system enforces them: the client is read-only and ignores code the agent writes, the sandbox sees only the software it runs, and its only network path is a logged relay to the model's API. For licensed data the daemon is also the licensing boundary: the data stays behind it, and agents, including hosted models, receive only what it discloses, derived statistics in the backtestable track, so a vendor's terms on processing and redistribution are met at one enforced point. The thesis track's daemon serves only public data. Even so, the ledger is an audit mechanism, not a measure of information leakage. It records what was disclosed, not how much an agent inferred from it, and it cannot see what the agent already knew from training.

4 A typed strategy language

Agents do not write arbitrary Python backtests. They write strategies in a small domain-specific language designed to rule out specified classes of research bugs before anything runs. Strategies are data, an abstract syntax tree that the kernel type-checks and interprets, never host-language code. The properties below are design targets; each must be stated as a formal invariant and tested, including against strategies written specifically to evade it, and the core ones are candidates for proof (Section 6).

Causality in data and in execution. A strategy is a causal stream transformer, formally a Mealy machine: its output at time t depends only on inputs it has received. That alone is not enough. A type signature only describes an interface; the guarantee comes from the runtime feeding the machine only point-in-time data and enforcing four distinct times: when data became available, when the strategy decided, when its order could be submitted, and when that order could actually fill. A strategy that reads a candle's close at 12:00 and fills at that close is causal in its inputs but still leaks through execution, and timestamps alone cannot exclude it, since such a fill can satisfy every time ordering with equality. Even a correct fill rule cannot establish that a fill would really have happened: queue position, cancellations, competing orders, and latency are not fully visible in historical data, so fills are an estimate from a calibrated execution model. Event-driven engines such as QuantConnect's LEAN already separate a time frontier from fill models, and the DSL should at least match that.

```
-- A strategy consumes point-in-time observations and emits orders
type Strategy = Mealy (Obs Event) Order

-- Each observation records when it became knowable and which revision it is
data Obs a = Obs { eventTime :: Time, knownAt :: Time, revision :: Int, value :: a }

-- The simulator, not the strategy, decides submission and fill times
```

```
data Fill = Fill { decidedAt :: Time, submittedAt :: Time
                 , filledAt :: Time, price :: Price }
```

Every event therefore carries a sequence number as well as a time, and the evaluator, not these types, enforces two conditions. For every input to a decision,

$$t_{\text{known}} \le t_{\text{decision}} \le t_{\text{submission}} \le t_{\text{fill}}, \tag{1}$$

and the fill is computed only from market events that occur strictly after the order became eligible to execute, ordered by time and then sequence number, so that events sharing a timestamp still have a definite order. Any exception, such as an auction that crosses at a price set after submission, is modeled explicitly.

In the pilot, a decision using the close at B occurs at $B + \delta$ and fills at the opening reference at $B + 4$ hours (Figure 2); Appendix A.2 derives that delay from the chosen bar grid.

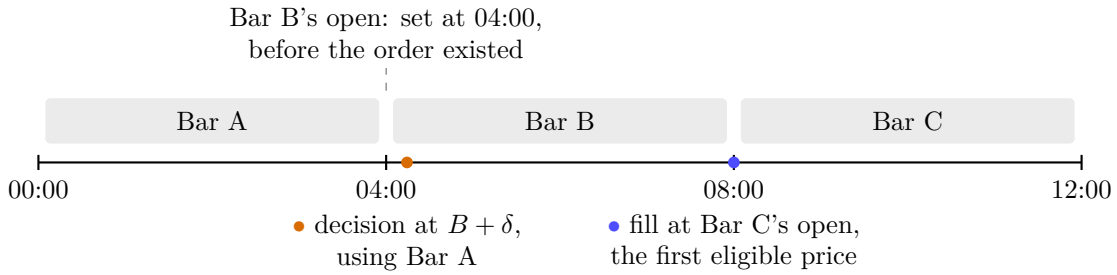


Figure 2: Fill timing in the pilot, with $B = 04{:}00$. Bar A closes and Bar B opens at the same instant, so Bar B's opening price predates the decision and is never used.

Units, dimensions, and time bases. Prices, sizes, funding rates, and volatilities carry types, so adding a price to a funding rate fails to compile. Rates also carry their time basis: an hourly and an annualized funding rate are both dimensionless ratios, but combining them without an explicit conversion is a type error. Normalized features get their own dimensionless type. Types do not stop a feature from being normalized with future data; that is the runtime's job.

Levels and changes. Many economic series trend, and two independent trending series routinely show large, meaningless correlations [12]. The type system distinguishes levels from changes. Correlation, regression, and ranking operators are to accept only changes, ratios, or other transforms declared stationary; applying them to levels is a type error unless the strategy explicitly invokes a cointegration test, whose result is a logged statistical finding rather than a license. This is planned with the kernel's generalization (stage 8); the pilot's DSL has no correlation or regression operators, and its ranking operator accepts any dimensionless value. It enforces declared transformation rules, not stationarity itself: a difference or ratio need not be stationary, and some sound models work on levels. It still removes the most common source of spurious cross-series findings, which matters most for open-ended exploration (Section 10).

Composition. Causal transformers compose: features feed signals, signals feed sizing, sizing feeds risk limits. With objects, morphisms, and parallel composition specified precisely, this should form a symmetric monoidal category, which would give the canonicalizer in Section 7 algebraic laws to rewrite with. That structure is a construction to carry out, not yet a result, and each rewrite law

must be shown to preserve full trading semantics, including state, order timing, and fees. Laws are proved over an abstract number structure; agreement in binary64 arithmetic is checked bit for bit by differential tests, not proved.

Execution realism. Fees, funding payments, slippage, and latency are part of the evaluator's semantics rather than optional parameters. Each venue and market class needs its own model: on Hyperliquid, the fee tier a small account actually pays, funding schedules, and order-book depth (the pilot fits one constant cost per coin, side, and size bucket to a 24-hour recording of order books, so depth at the decision time is not modeled); for builder-deployed markets, also the market's oracle and trading-hours behavior (Section 11). These models are only as good as their calibration, so the paper treats them as validated components with stated error bounds, not as solved. Where calibration data is thin, for example where historical order books do not exist, the limitation is reported alongside every result that depends on it (Section 8).

5 Sources: declaration, certification, integration

A strategy should be able to say "this depends on a feed of shipping disruptions" before anyone has built that feed. The DSL therefore treats a data source as a typed declaration, separate from any concrete feed.

Declaration. A source declares a schema: an entity key, typed fields with units, a cadence or event semantics, and an availability policy. The `knownAt` time of every observation is assigned by the recorder or the certified backfill, never by the feed itself.

```
source ShippingAlerts : EventStream {
  key      : Region
  kind     : Enum [Closure, Delay, Attack]
  count    : Count
}  -- knownAt is assigned by the recorder, never by the feed

strategy S (alerts : ShippingAlerts, brent : Price Commodity) = ...

bind ShippingAlerts := feed "alerts-v3"   -- refused without a valid certificate
```

The type checker checks a strategy against the declared schema without any binding. Evaluation requires every source the strategy names to be bound to a certified feed; an unbound source makes evaluation undefined, not empty. Agents can therefore propose hypotheses about data that does not yet exist, and those hypotheses wait in the ledger until the feed is certified. An unbound declaration prevents fitting through this system before the feed binds, but it does not make the strategy a pre-registered prediction: the agent may already know, from training data, published research, or an equivalent feed, how the relationship behaved. Only a frozen strategy evaluated on data independent of its selection supplies independent evidence (Section 8).

Certification. A feed binds to a declaration only with a certificate recorded in the ledger. The certificate holds the results of tests with thresholds fixed before the feed's data is examined against any returns:

- **Terms:** the source's terms permit the automated use: the certificate quotes the clause that permits it, with its URL and retrieval date, or shows that the feed is an official API whose

terms allow it. Silence is not permission. A source whose terms forbid automated access is never fetched, and anything collected from it before that was found is deleted. Where the terms restrict the operator rather than the access, as some venues' interface terms restrict residents of certain jurisdictions, using the read-only public data is the operator's decision, recorded with the clauses found.

- **Availability:** every historical timestamp is classified as directly observed (recorded live, or witnessed by an independent archive capture or a protocol-enforced time), reconstructed from the source's own release records, or estimated. A lag measured on a live recording window validates the availability rule going forward; applied to history it is an assumption, since lags change over time. A historical row's known-at time is therefore the later of the claimed-time rule, used only where the claimed time is protocol-enforced or server-set and the live test passed, and any independent witness. A row with neither is usable only prospectively.
- **Revisions:** a later re-fetch of the same history compared against what was recorded live, so that a backtest never sees corrected values nobody had at the time.
- **Entity mapping:** the precision and recall of the mapping from raw records to schema keys, measured on blind-labelled samples, with thresholds justified by how much mapping error would dilute the signals that use the source.
- **Coverage:** gaps, outages, and the period over which the feed is complete.
- **Extractor validation** for feeds derived by an LLM (below).

Agents may build feeds, but the kernel's certification tests decide whether a feed binds, so an agent never grades its own data. The pilot's mapping cutoff is a post-measurement choice, not a derived dilution bound (Appendix A.4).

Acquisition. Free data comes first, and the data layer is designed for extension: adding a feed is a declaration, a terms record, and a certificate, never a code change. Official sources are terms-certified and recorded first, keyless or keyed (a key or account is registered in the operator's name); the full certificate, with the tests above, comes with source declarations (stage 7). Paid data is the operator's decision: the candidates are a survivorship-free equity history with delisted securities and historical index constituents, and exchange futures histories with contract definitions and settlements, each subject to the licensing boundary (Section 11). On-chain data is a priority family: chain data read from one's own node or an official archive carries no data provider's terms, though the access route (a cloud account, a requester-pays archive, an RPC provider) has terms and costs of its own. Where a source keeps no past versions, our own recording is its only point-in-time record, so recording starts early and runs on more than one host. Each host's files are pulled to the primary copy, never pushed, so losing a host loses nothing already pulled. A source's results move up a claim tier (Section 11) as better data arrives.

Derived and LLM-extracted sources. A derived source is a DSL function of other sources, available no earlier than the latest of its inputs plus its processing time. When an LLM turns raw text into structured events, it is restricted to factual fields: who, what kind of event, and when. Judgments such as sentiment or importance are excluded, because a model that knows how events turned out can shade those judgments with hindsight. The extractor is validated with names and dates masked, its outputs are cached with the model version and prompt hash, and that version is part of every experiment's identity. Feed design choices, such as which keywords define an event, are

themselves forking paths, so the feed specification is pre-registered before any returns conditioned on it are examined.

Schemas as categories. In Spivak's framework, a database schema is a category and an instance is a set-valued functor on it. Objects are entity types such as Asset, Issuer, Event, and Post; morphisms are functional relationships. Many-to-many, time-varying relationships need their own entities: a wallet does not map to one token, so the schema has a Position object with morphisms to Wallet, Token, and an interval of validity. In these terms a source declaration is a schema, a certified feed is an instance of it, and a strategy written against the schema accepts any certified instance. Ologs are the human-readable version of the same idea.

Migration as functors. Functorial data migration, implemented in the CQL tool [5], moves data between schemas along functors and preserves the declared relationships. It does not solve entity resolution: if two sources use different identifiers for the same asset, or the same identifier for different contracts, a migration can be formally correct and still wrong, which is why entity mapping is a certification test. An open design question is whether this framework earns its complexity over a carefully typed, versioned relational schema; its main advantage would be that migrations and their consistency properties are explicit and checkable.

Point-in-time data. A `knownAt` field is necessary but not sufficient. Real data is revised, backfilled, delayed by ingestion, and sometimes mistimestamped. The store keeps immutable revisions with lineage. A query "as of time t" resolves each key to the latest revision fetched by the dataset version and shows it only if its `knownAt` is at most t; it never falls back to an older revision, so a later correction hides a value rather than replacing it with one nobody had, and the revisions test measures what that costs. A revision stores only what was observed: the values, the source's own timestamps, and when it was fetched. Its `knownAt` is computed from those fields by the availability rule that the dataset version pins, so a corrected rule reaches every revision at once, and no output of a superseded rule is ever read. Some public archives already work this way: ALFRED [1] serves U.S. macroeconomic series as each vintage was published. Its vintages are dated by day and are typically added within a business day of release, so they establish which value was current on a date, not when it became tradable within that day; intraday release times come from a separate release calendar, certified like any other source.

Cross-source hypotheses. Agents can propose strategies as paths through the schema, for example from a central-bank statement, to the currencies and rate-sensitive sectors it affects, to the commodities those sectors consume. Each such path is a hypothesis recorded in the ledger. A schema with many sources makes it easy to generate many plausible-sounding paths, which is why exploration across sources has its own design (Section 10).

6 Verified reference kernels

The DSL's semantics, type checker, and canonicalization rules are the parts of the kernel whose correctness can be proved rather than only tested. The reference kernel is a Lean 4 [9] model of them, in which the theorems below are proved. It is differentially tested against the production engine, and the two must agree exactly. That engine is temporary: an executable Lean engine, with OCaml for untrusted plumbing, replaces it and the Python around it, under that step's own criteria, which include matching the engine bit for bit on the frozen corpus. A second candidate was

evaluated against pre-registered criteria and removed. Independence already comes from the engine itself and from components implemented separately by a model from another lab, so a second proof kernel kept in lockstep would add maintenance more than assurance.

Theorems. Six statements, pre-registered before any proof is attempted:

1. **Non-interference (causality).** For every well-typed strategy s, any two input histories that agree on every observation known by time t produce the same decision at t:

$$h_1|_{\text{known}\leq t} = h_2|_{\text{known}\leq t} \implies \text{decide}_s(h_1, t) = \text{decide}_s(h_2, t). \tag{2}$$

2. **Unit soundness.** Every well-typed expression evaluates to a value of its declared unit and time basis.

3. **Rewrite soundness.** Each canonicalization rule preserves the strategy's full trading semantics, including state and order timing.

4. **Fill ordering.** Every fill satisfies both timing conditions of Section 4: the time ordering, and a fill price computed only from market events strictly after the order became eligible, in time and sequence order.

5. **Binding.** Evaluation is undefined for any strategy that names an unbound source, and, once certificates exist in the kernel (stage 7), an uncertified one.

6. **Bounded returns.** Losses are capped at account equity, so every reported per-step return is at least -1.

Numbers are kept abstract in the proofs. Proving properties of IEEE floating-point code is possible but costly, and several algebraic identities, such as associativity of addition, are false for floats. Rewrite rules are therefore either restricted to float-safe identities, as the pilot's canonicalizer already does, or proved over an abstract ordered field, with the engine's numerical differences bounded by differential tests at a stated tolerance.

What proofs reach. Proofs establish properties of the formal semantics, which is only the first of three levels of assurance (Table 2).

Level	What it establishes	How
Language	Causality, unit soundness, rewrite soundness, fill ordering, binding, and bounded returns in the formal semantics	Proofs (this section)
Implementation	The production engine computes what the formal semantics specify	Differential testing now; a refinement proof, as in verified compilers [21], is future work
Economic model	Simulated fills, costs, margin, and funding approximate the real venue	Empirical calibration only

Table 2: Levels of assurance.

A simulator can satisfy its fill-ordering theorem while charging the wrong fee schedule or misjudging available liquidity, and two implementations that agree can share the same wrong model of the market.

The laws are the new spec. A proof checker certifies that a proof matches its statement, not that the statement says what was intended. If the same agent writes both the law and the proof, it can state a weak law and prove it. The laws are therefore treated like stage criteria: written first, independently reviewed by a model from a different lab, frozen, and never weakened afterwards. A law that turns out to be wrong gets a dated revision, and proofs of the old statement are not counted toward the new one.

7 Canonicalization and the experiment ledger

The ledger makes the research process auditable; it does not by itself produce a trial count that can be plugged into a correction. A canonical strategy is a program, while a statistical trial is about information: what an agent saw and how that shaped what it selected next. An agent that inspects one strategy's drawdown, yearly returns, and cost sensitivity, then proposes something new, has spent multiplicity without creating any new canonical form. Conversely, rerunning an identical deterministic backtest, or rejecting a strategy at the type checker before it touches data, spends none.

What the ledger records. Six things, kept distinct:

1. **Strategy identity:** the canonical form, so equivalent programs are recognized.
2. **Experiment identity:** strategy, parameters, asset universe, data window, dataset and certificate versions, availability-time conventions, execution and fee model versions, extractor versions, simulator version, and random seeds.
3. **Disclosures:** every statistic or plot returned to an agent, including for rejected candidates.
4. **Selection decisions:** which candidates an agent or the gate chose to pursue, and what it had seen at that point.
5. **Source certificates:** the tests and thresholds under which each feed was bound.
6. **Theses:** registered forward predictions and their retrieval logs (Section 12).

Multiplicity is then estimated from this history, and those estimates are checked by testing the whole procedure on null markets (Section 16).

Canonical forms. Strategies are rewritten using the DSL's algebraic laws, for example commuting independent features or folding constant transforms. Equality saturation with e-graphs, as in the egg library, is a practical tool for this, with an important limit: it finds expressions provably equal under the supplied rewrite rules and extracts one by a cost model. It does not produce a unique canonical form for every semantically equivalent program, and results depend on rule coverage and resource limits. The claim is therefore narrower: the system detects some provable equivalences, conditional on the soundness of its rewrite rules, which Section 6 aims to prove.

Effective N. Strategies that differ syntactically can still behave almost identically. Clustering return streams by correlation gives one estimate of the effective number of independent trials, in the spirit of Bailey and López de Prado, but it is a heuristic. The correct adjustment depends on the joint distribution of the test statistics, and the correlations are estimated on the same sample being searched. The gate therefore reports how its decisions change across several effective-N estimates, and treats a promotion that holds only under the most favorable estimate as fragile.

Search guidance. Agents can query which regions of strategy space are already explored, steering generation toward novel hypotheses. This is itself a feedback channel: telling agents where past results were good or bad is a disclosure, and it is logged and accounted for like any other.

8 The statistical gate

Individually valid tests do not make a valid combined procedure. The gate therefore separates selection from inference. Everything computed on the search period, including the screen, SPA, Romano–Wolf, and the mechanism check, is a selection rule: it decides which candidates reach the holdout, and its behavior is calibrated by simulation (Section 16), but it supports no error claim on its own. Romano–Wolf's guarantees, for example, do not survive applying it to a family that was itself chosen on the same data. The only inferential step is the holdout batch: finalists frozen without any holdout data are tested once on that untouched period, with Holm's correction across them. Each campaign runs a fixed gate written down before it starts, and its error rate is claimed only for its finalist family on its holdout, under the holdout test's scope assumption that every net return before the venue absorbs a shortfall is at least -2.

1. **Search period only.** Agents see results from a research period. The holdout period is never shown to them.

2. **Screen.** Candidates must pass a deflated Sharpe threshold computed at each of several effective-N estimates, using the largest. That is conservative only relative to the estimators chosen, not a guaranteed bound.

3. **Global screen.** Hansen's SPA runs over every strategy evaluated in the campaign and asks whether any beats the benchmark. If it does not reject, the campaign promotes nothing, which under the complete null stops most campaigns before any holdout is spent.

4. **Ranking.** Romano–Wolf stepdown ranks the screened candidates by adjusted p-value, and those it rejects at the family level become finalists. These p-values are used for selection only, since the screened family was chosen on the same data.

5. **Mechanism check.** A candidate whose pre-registered mechanism prediction fails is removed from the finalists. There is no override by a higher statistical threshold.

6. **Freeze.** The finalists, the promotion rule, and the multiple-comparison correction across finalists are fixed before the holdout is opened.

7. **Holdout batch.** All finalists are evaluated together, once. The kernel enforces this: it reads the whole batch before evaluating any of it, accepts it only if its trials are exactly the finalists frozen in the ledger, and claims the window under a lock before the first evaluation. A finalist is promoted to paper trading only if its net-of-cost holdout return is positive and its holdout p-value for the test below passes a preset level after Holm correction across finalists. The thresholds are parameters; the structure is fixed.

8. **No recycling.** A strategy modified after its holdout result cannot be retested on that holdout. A holdout used by one campaign is spent: the next campaign evaluates on a later period, and the old holdout becomes part of its search data.

The holdout test. For each finalist i, the null hypothesis is that at every step of the holdout its net return, before the venue absorbs any shortfall, has non-positive mean given everything before that step, which includes everything the search revealed:

$$H_{0,i}: \ \mathbb{E}\big[r_{i,t}^{\text{net}} \mid \mathcal{F}_{t-1}\big] \le 0 \quad \text{for every } t, \qquad \mathcal{F}_{t-1} \supseteq \mathcal{F}_{\text{search}}. \tag{3}$$

The gate uses a predictable, capped plug-in Kelly bet and the inverse running maximum of its wealth as a p-value. Appendix A.1 derives conditional validity and Holm's guarantee, including on bankrupt paths, under the pre-absorption lower bound -2. That bound is proved only for the corrected gapless synthetic construction (Appendix A.2), not for the frozen pilot or real markets. Reported returns' floor of -1 is not a substitute. A locked null suite checks the implementation, not the theorem's premises in every market. This bet has very low power at the pilot's holdout lengths (Appendix A.1); zero promotions is not evidence of no edge.

Calibration and scale. Cheap baseline searchers supply the large campaign samples; expensive agent campaigns are smoke tests, with much weaker bounds. Appendix A.1 derives their confidence bounds and distinguishes the chosen targets from evidence of useful sensitivity.

Across campaigns. Error control is per campaign (per arm in stage 3), not project-wide. Every promotion is reported beside the number of campaigns. A project-wide claim needs a precommitted summable error budget, derived in Appendix A.1; alpha-investing [11] controls a different quantity and is not a substitute. Successive holdouts are later periods, never recycled.

Reported, not judged. Beside each finalist's judged holdout result, the gate reports its holdout return at twice the modeled costs and the share of the holdout period whose market conditions fall within the range the cost model was calibrated on. These reports appear only after the promotion decision and are never used to choose among finalists. They exist because cost models calibrated on short windows may understate costs exactly when markets move and strategies trade most.

Method	What it controls	Key assumption to check
Deflated Sharpe ratio	Selection bias from N trials, non-normal returns, track length	A valid effective N and the Sharpe sampling model
White's Reality Check, Hansen's SPA	Whether the best of a family beats a benchmark	A correctly specified family and a suitable bootstrap for dependent data
Romano–Wolf stepdown	Family-wise error across correlated strategies	Resampling assumptions; it controls error, it does not certify survivors
Benjamini–Hochberg [6]	False discovery rate	Valid p-values and acceptable dependence
Probability of backtest overfitting (CSCV)	Instability of the in-sample selection procedure	A diagnostic of the procedure, not a forecast of future profit

Table 3: Statistical methods used by or available to the gate.

The DSR is a selection heuristic, not inference after adaptive search. Its per-period formula, reference scale, numerical conventions and limitations are derived in Appendix A.1; simulation measures the complete procedure only on the specified generator families.

The holdout. Every query to a holdout leaks some information, and what matters is how much each answer reveals, not just how many queries there are. The reusable holdout of Dwork et al. (Thresholdout) releases holdout answers through a noisy threshold mechanism and gives formal guarantees under adaptive queries. If Thresholdout is not implemented, the system's hard query budget is a heuristic safeguard only, and the paper does not claim formal adaptive validity for it.

9 Mechanism obligations

Every strategy must ship with a stated mechanism, and the mechanism must imply tests beyond PnL. These tests are additional empirical checks that can count against a strategy; they are not a second verifier comparable to a proof checker. The obligation has three parts:

1. **Counterparty.** Who is on the other side of the trade, and why are they willing to trade at this price? The counterparty need not be wrong: forced sellers at liquidation, hedgers paying for immediacy, index funds rebalancing on a schedule, and traders reducing inventory can all be acting rationally.

2. **Conditional predictions.** If the mechanism is real, the effect should vary in specified ways, with stated tolerances. A liquidation-cascade strategy should earn more when positions are crowded near liquidation and less when funding has normalized. A slow-information strategy should decay as the information ages.

3. **Capacity and decay.** Roughly how much size the effect supports, and what would make it disappear.

Conditional predictions are written down before they are tested and logged in the ledger like any other experiment. Ideally they are tested on data not used to find the strategy, since predictions checked on the same sample are not independent of the PnL result. A failed prediction is evidence that the stated explanation is wrong, not proof that the strategy is: it may work for another reason, or the test may lack power. Under the gate in Section 8, a failed pre-registered prediction removes the candidate from that campaign's finalists; it can be re-proposed with a revised mechanism in a later campaign, evaluated on fresh data.

A working hypothesis of this design is that LLMs are most useful here for proposing economically plausible ideas rather than for blind search. That hypothesis should be tested against simpler generators, such as random or genetic search over the same DSL, using the same gate.

10 Exploration across domains

For a retail researcher, "alternative data" cannot mean buying proprietary feeds. It can mean using agents to look for relationships between sources that no human would think to combine, for example shipping disruption reports, regional electricity prices, and a basket of industrial equities. This is also the statistically most dangerous activity in the system: the space of cross-source combinations is effectively unbounded, so by Table 1 the best in-sample result of an open-ended exploration is uninformative on its own, however striking it looks.

The design therefore uses the same split as the gate, the classical data-splitting approach [8], with two changes for open-ended search:

1. **Free exploration.** On the search period agents search without any screen. Every evaluation is still logged, but nothing found counts as evidence; a screen at the exploration's N would reject everything.

2. **A small, fixed number of finalists.** Agents promote at most k hypotheses, a number fixed in advance, each with a mechanism and with its parameters, source definitions, and asset selection frozen.

3. **Confirmation on untouched data.** The k finalists are tested once on a later period, separated from the search period by an embargo gap, with Holm's correction across k. The multiplicity paid is k, not the size of the exploration, which is what makes the design affordable. Where history allows, an intermediate confirmation period can precede the final holdout, so that only hypotheses confirmed once reach it.

The qualification is independence. An embargo reduces serial dependence between periods but does not make later markets independent of earlier ones, and any decision made after looking at confirmation data invalidates it. Splitting also costs data: each period is unavailable to the others, and each confirmation period, like a holdout, is spent once used. The design is therefore practical only on long histories, which is a further reason for the multi-asset scope. The level-versus-change types (Section 4) guard against the most common spurious cross-series finding, and the mechanism obligation applies with particular force: a relationship between unrelated domains must name the economic channel that links them, or it does not reach confirmation.

An honest caveat: a relationship nobody would think to trade may persist precisely because nobody trades it, but it is far more likely to be noise. The expected outcome of most exploration campaigns is zero confirmed hypotheses.

11 Multi-asset research and HIP-3 markets

Hyperliquid's HIP-3 standard lets third-party builders deploy perpetual futures markets [18], and builder-deployed markets on equities, equity indices, and commodities have grown quickly. This makes a single venue a plausible execution layer for many asset classes, for operators who may use it. The research layer, however, should not use those markets' own histories, which are months long. It uses the long histories of the underlying assets, and treats each venue market as an execution mapping that must be certified separately.

Research layer. Each asset class needs point-in-time data:

- **Equities:** prices adjusted for splits and dividends, and a universe that includes delisted companies. Freely available equity data usually omits delisted names, which biases every backtest upward. Survivorship-free sources such as CRSP exist but carry licensing terms that must permit the intended use.
- **Indices:** index levels and, for constituent strategies, point-in-time membership.
- **Commodities:** futures series with explicit, versioned roll rules, since the choice of roll method changes returns.
- **Macro and rates:** vintage-correct releases (Section 5) and a calendar of scheduled events with their release times.

Universe selection is itself a lookahead. The set of assets that have HIP-3 markets today was chosen in 2025 and 2026, largely by popularity, and popularity reflects past performance. A twenty-year backtest restricted to today's listed stocks is a backtest on yesterday's winners. The

research universe must therefore be defined point-in-time, for example by index membership as it stood on each date, and the list of currently listed venue markets is used only to map research positions to execution.

Execution mapping. Before a research result on an underlying asset can be traded through a venue market, that market is certified like a source:

- **Tracking:** the gap between the venue's mark price and the underlying, measured over a recording window, including around corporate actions such as dividends and splits, whose handling may differ by deployer.
- **Closed hours:** the perpetual trades continuously while many underlying markets close overnight and at weekends. Price discovery then happens on the venue alone, and reopening gaps against the underlying must be measured, not assumed.
- **Funding and costs:** the funding mechanism and realistic costs for the account size.
- **Market risk:** deployer-supplied oracles and parameters, and the possibility that a builder market is halted or delisted.

A strategy whose edge depends on the closed-hours behavior of a venue market cannot be researched on the underlying at all, because the underlying has no prices in those hours; it falls back to the venue's own short history and the corresponding statistical penalty.

Underlying research and venue claims. Research may establish either benchmark predictability or an executable backtest on an underlying instrument. Neither establishes a backtest of a different venue market, whose returns also include funding, basis changes against the underlying, mark-price dynamics, liquidation risk, and costs, any of which can change the sign of an edge. A projection to that market uses a separately certified execution mapping and remains labelled model-dependent until supported by the market's own forward record.

Two tiers of underlying claim. Free data often covers a benchmark series (a reference fixing, a policy rate, a monthly commodity benchmark) long before it covers an executable instrument. A *signal study* claims only that a signal predicted a named benchmark under a predeclared target-resolution and scoring rule; it makes no execution, cost, capacity or venue claim. Its inputs and target endpoints require certified point-in-time evidence, so a source without recoverable, evidenced past versions is usable only prospectively, and its exact score must satisfy the statistical gate's lower-bound and conditional-null assumptions. An *executable backtest* additionally requires executable instrument histories and the applicable corporate-action, contract, roll, financing and cost mechanics. Both tiers keep their calibration, information-boundary and holdout requirements. A source moves up a tier only when better data arrives, and moving up requires newly preregistered evidence and fresh adjudication; an earlier signal-study result does not satisfy the stronger claim.

Licensing boundary. Some sources' terms restrict use with AI or language-model systems while permitting storage and deterministic processing: exchange data such as CME's, vendors such as Norgate, and Nasdaq-originated data. The role of each source is a field of its declaration, set from its terms record and checked by certification: *agent-readable* only where the record quotes a grant covering reuse, *kernel-only* where it quotes storage and deterministic processing, and excluded where even that is forbidden. No vendor is asked for written permission. A kernel-only source's

rows enter only deterministic code (ingestion, certification tests, kernel and gate). Research agents and reviewers see only its permitted reports, and the operator's coding sessions see at most small amounts while debugging, preferring counts, hashes and test results.

- **What may cross:** the fields of a frozen per-source report list, each with its minimum aggregation stated as a count of decision dates and instruments per reported cell: for example results by year and by regime, contribution by instrument and class, turnover, exposure, hit rate, results split by the strategy's own condition, and drawdown timing to the calendar month. Schemas and metadata cross only where the terms do not restrict them.
- **What never crosses:** any per-date or per-row value, an equity curve finer than a calendar year (the curve of a long-only position in one instrument is its price series), and any cell below its minimum aggregation.

Evaluation windows on a kernel-only source are fixed by the class annex, never chosen by an agent, and its strategies may not condition on calendar dates. The boundary belongs to the disclosure path (Section 7), and certification runs a frozen attack suite against it: single-date conditions, one-date differences between two strategies, long-only single-instrument strategies, and threshold bisection on a feature. It passes only if no per-date value is recovered within a stated tolerance from any combination of reports within a campaign's evaluation budget. Where a vendor requires its data deleted when an account ends, the audit record keeps hashes, schemas, certificates and kernel outputs, never the vendor's rows, so far as the vendor's terms allow; a kernel-only result can be replayed only while subscribed.

Venue eligibility. Eligibility is part of feasibility, not a final step. Hyperliquid's interface restricts some jurisdictions, including the United States, so an operator in one of them cannot assume access to its markets. A research result on an underlying asset is portable: the execution mapping can target any venue the operator may use, including regulated instruments on the same underlyings, and the certification steps above then apply to that venue instead.

Breadth from a pilot. Stages 0 to 5 of the roadmap build and test the machinery on crypto perpetuals. Extending to these asset classes keeps the architecture: the kernel's role, the gate, and the ledger. The present kernel, however, models only a perpetual that trades around the clock, with mandatory funding and a fixed 4-hour decision grid. Stage 8 therefore generalizes it to instrument kinds with their own calendars, settlement, and decision cadences, alongside new sources, certificates, execution mappings, and null models whose cross-asset dependence matches the larger universe.

12 The thesis track: event-driven macro

The research direction we consider most promising for an agent-assisted retail trader is discretionary, event-driven macro: theses about how a scheduled or unscheduled event, such as a central-bank decision, an inflation release, an earnings report, a supply disruption, or a regulatory decision, will move a specific exposure over days to weeks. This is where reading and synthesis matter most and where speed matters least.

Why this track is forward-only. An LLM's judgment cannot be reliably backtested on a period its training data may cover: a model reading news from 2024 may already know what happened next, even if indirectly. A frozen model can be evaluated on a period that begins after its training

cutoff, but cutoffs are imprecise and later fine-tuning can leak later information, so the margin must be generous and the usable period is short. The pilot's single-agent campaign therefore uses a short historical holdout one month past its model's cutoff as a pipeline validation run, not a test for edge, and anything it promotes is confirmed on a later forward window. LLM forecasting research reaches the same conclusion and evaluates on questions resolved after the model's cutoff [14, 20]. Theses are therefore registered before their outcomes are known and evaluated only forward.

Registration. A thesis is a structured record, hashed into the ledger at registration and never edited:

```
thesis T-0142 {
  registered   : set by the kernel at write time
  exposure     : long crude oil vs. short airline basket
  horizon      : resolves at a fixed date, three weeks out
  claim        : P(exposure return > 0 over the horizon) = 0.64
  catalyst     : a scheduled producer meeting; reported inventory draws
  invalidation : close early if the exposure loses more than a stated amount
  sources      : ids in the retrieval log
  size         : fixed risk budget / distance to invalidation
}
```

The retrieval log records every document read, not the model's training knowledge. The daemon sizes positions and caps nominal reservations at admission; realized losses are not capped by those budgets. Appendix A.5 derives the sizing and explains the choices. Registration atomically records its time, size, reservation, cost cells and base rate; replay consumes those records, never recomputed inputs. Invalidation is predeclared; any discretionary exit is a separate decision, with position performance reported both as registered and as managed so exits cannot hide losers. The probability is scored, not used for sizing.

Prices. Positions are marked with the venue's public data. A builder-deployed (HIP-3) market's trade candles have hours without trades, so it is marked instead at its sampled mark price: one sample shortly after each hour, sealed five minutes later and never changed afterwards. A price informs a decision only once it is available. A fill is never priced from an earlier price: an exit moves to the next available reference, and a thesis with no reference between its entry and its horizon is unresolved. Funding, an accounting entry rather than a decision, is valued at the latest earlier reference when the hour's own is missing.

Scoring. Probabilities receive Brier scores against an exposure-specific historical comparator and a fresh conditional no-reading ablation. A full no-reading agent and a frozen benchmark policy are scored separately; named outcome-market prices are descriptive only. Position performance uses the fixed sizing rule. Appendix A.5 gives the score derivations and why these comparisons answer different questions. Graph clusters proxy dependence; counting them does not establish independence.

Sample size. Forward evaluation is honest but slow. Appendix A.5 derives illustrative independent-hit-rate sample sizes, not a power calculation for clustered paired Brier skill. Stage completion is operational; scores remain descriptive without a separately preregistered inferential protocol. Starting early accumulates evidence, not permission to claim skill early.

Connection to the backtestable track. The thesis agent may use the backtestable track's certified sources and promoted strategies as inputs. In the other direction, the registered theses form a point-in-time record whose `knownAt` is the registration time, so once enough have accumulated, the thesis log itself can be certified as a source for the backtestable track.

13 Research directions for retail edge

The method is not the edge: institutions already run automated alpha research. The directions below are hypotheses about where a small account might compete, not demonstrated sources of return.

- **Event-driven macro theses** (Section 12). Large funds run discretionary macro and event-driven books too, but at small size an agent-assisted researcher can cover many minor catalysts that do not justify an institutional analyst's time.
- **Broad, known premia.** Searches for novel technical patterns rarely survive correction for data snooping [29]; the technical effects with long records are broad premia such as time-series momentum [25, 17], which work through diversification across many markets. These are also available cheaply through funds, so a homemade version must beat them after costs.
- **Cross-domain relationships** found through split-sample exploration (Section 10).
- **Small capacity.** Some strategies support too little size to interest a large fund, though institutions do not ignore small opportunities uniformly.
- **Venue structure on Hyperliquid.** Positions and margin for given addresses are far more observable than on centralized venues, which makes crowding, deleveraging, and funding dislocations testable in principle. In practice, liquidation prices depend on cross-margin account state, funding paid, and partial-liquidation rules, so they cannot be rebuilt from entry price and leverage alone, and historical archives can have gaps.
- **Slow, messy information.** Governance proposals, filings, and long-tail discussion may be absorbed by the market slowly; LLMs make reading them cheap, though they make it cheap for competitors too.

Observed wallet activity is treated as data rather than something to copy directly. Among thousands of visible wallets, many strong records are luck, which deflated statistics can partly correct. Survivorship is a different problem: wallets that blew up or were abandoned may be missing from whatever set looks visible today, and no adjustment to the survivors' Sharpe ratios recovers them.

14 Related work

Agentic research automation itself is not new. The possible contribution is narrower: combining a restricted strategy language with proved core invariants, typed and certified sources, a full record of what agents saw, point-in-time evaluation, multiplicity-aware inference validated by null simulation, and a forward-only thesis track in one ledger. Establishing that this combination adds something requires a direct comparison with the systems below, which this draft does not yet make.

- **Automated quant research.** Microsoft's R&D-Agent-Quant [22] already runs a closed loop of hypothesis generation, code development, market evaluation, and feedback. Alpha-GPT

[31] uses LLMs to mine alpha factors interactively. Kakushadze's 101 Formulaic Alphas [19] publishes formulaic alphas used in industry.

- **Backtesting infrastructure.** Event-driven engines such as QuantConnect's LEAN [24] already use a time frontier to limit future-data access and separate fee and fill models. The DSL here must offer stronger, stated guarantees to justify itself.
- **Multiple testing in finance.** Harvey, Liu and Zhu [16] argue for a t-statistic hurdle near 3 for new factors. Bailey and López de Prado introduce the deflated Sharpe ratio [4], and with Borwein and Zhu the probability of backtest overfitting [3]. White [34], Hansen [15], and Romano and Wolf [26] provide bootstrap tests for data snooping, and Sullivan, Timmermann and White [29] apply them to technical trading rules.
- **Adaptive data analysis, data splitting, and sequential testing.** Dwork et al. [10] show how to reuse a holdout set under adaptive queries with formal guarantees, for independent samples rather than dependent time series; Cox [8] analyses the classical split-sample design used by the gate; Foster and Stine [11] give alpha-investing for sequences of tests.
- **Forecasting.** Brier [7] introduced the probability score used for theses; Tetlock and Gardner [30] document what distinguishes skilled human forecasters; recent work evaluates LLM forecasters on questions resolved after their training cutoffs [14, 20].
- **Formal methods.** Equality saturation (egg) [35] supports detecting equivalent expressions; Lean 4 [9] provides the proof checker for the reference kernel; CompCert [21] shows what a refinement proof connecting formal semantics to an implementation involves.
- **Categorical data integration.** Spivak and collaborators develop ologs [28], functorial data migration [27], and the CQL tool [5].

15 Open problems and risks

The weakest link is non-stationarity: the inference methods in Section 8 rely, in different ways, on the sample being informative about the future, and markets drift and adapt.

- **Multiplicity under adaptivity.** Adaptive-inference theory exists, but no general method turns an agent swarm's interaction history into a trustworthy multiplicity estimate. Null simulation of the whole procedure is the practical check, and inference is confined to untouched holdouts.
- **Overfitting to the null simulator.** A gate refined until it passes its own null models may fail on others. Null models used while designing the gate are kept separate from locked evaluation suites, specified independently and run only to judge: in the pilot, a locked suite judges the holdout test; from stage 8, one judges the whole procedure.
- **Error across campaigns.** Per-campaign control does not bound the project's lifetime false-promotion count (Section 8).
- **Incomplete canonicalization.** Behavioral equivalence is undecidable for expressive languages, though decidable for finite-state Mealy machines over finite alphabets. A DSL with real-valued state falls on the hard side. Return-correlation clustering flags behaviorally similar strategies but is only a similarity heuristic.

- **Expressiveness trade-off.** A DSL strict enough to forbid leakage may forbid useful strategies. Extensions are acceptable only if they preserve the stated invariants and the proved theorems.
- **Proof-checker trust.** A soundness bug in the proof checker would vouch for leaky strategies. Lean 4's kernel is small and mature, its proofs can be re-checked by independently implemented kernels, and each proof's axioms are listed.
- **Hindsight in LLM-derived data.** Masked validation reduces, but does not eliminate, the chance that an extractor's outputs reflect what the model knows about later events.
- **Data quality and licensing.** Availability times for text sources are hard to establish after the fact, archives can be revised or incomplete, and survivorship-free data for traditional assets often carries licensing restrictions. Many sources' terms are silent on automated use, and silence is not permission.
- **Short venue histories.** Hyperliquid's core markets have a few years of data and its builder-deployed markets only months. Researching on underlying assets mitigates this only for edges that do not depend on venue-specific behavior.
- **Slow forward evidence.** The thesis track needs years to distinguish modest skill from luck (Section 12).
- **Venue and access risk.** Smart-contract, oracle, deployer, and liquidity risks can be partly modeled, but their severe tails are poorly represented in short samples. Live trading must use a venue the operator is permitted to access under its terms and applicable law, which is checked during feasibility (Section 11); the research pipeline itself is venue-agnostic.
- **Crowding.** If the approach spreads, the small-capacity edges it targets may shrink.

16 Roadmap

Check that the data can support the research before building the kernel, and build the kernel before the swarm. Evaluation criteria for each stage are fixed before the data they judge is inspected, so refining the evaluation does not become another layer of selection. A criterion that proves wrong gets a dated revision, and the stage is judged only on evidence produced after it. After a failure, a criterion may be amended only if the change cannot increase false discoveries, and the amendment is labelled as such, never as a pass of the original. A defect in the procedure itself that a failure exposes is fixed only by the human, as a new version judged on fresh evidence; the failure stays recorded, and the new version's result is never reported as a pass of the old. Stages 0 to 5 form a pilot on Hyperliquid's core crypto perpetuals; stages 6 to 10 extend it. Stages 6 and 10 depend only on the pilot's kernel and ledger and can run while the pilot's evidence accumulates. Stages 7 to 9 change the kernel, so their kernel changes wait until the pilot's judged runs are complete, and every judged pilot result comes from one frozen kernel. Each experiment records the kernel's build hash, so a change made before then would be visible in the ledger.

0. **Feasibility.** Pick one strategy family, for example funding dislocations, and confirm that every feature it needs was observable at each historical decision time and that realistic execution can be simulated. Version the venue's trading and margin rules alongside the data, since they can change over time. Exit when a point-in-time replay of that family is possible, or stop and pick another family.

1. **Kernel MVP.** A minimal DSL with stated invariants, a point-in-time evaluator with fee, funding, and fill models, and a ledger that records evaluations and disclosures. Exit when an adversarial test suite, including strategies written to evade the rules, fails as specified.

2. **Null, mixed, and power calibration.** First define "no edge" for each null model, for example zero expected net-of-cost return for every strategy expressible in the DSL. Preserving raw return autocorrelation contradicts this, since autocorrelation is itself exploitable; models such as martingale-difference processes with volatility clustering and cross-asset dependence keep the volatility structure without a predictable mean. Then run the entire adaptive search and gate on many simulated markets and measure three things against targets fixed in advance:

   - **Complete null:** the probability of promoting anything, with its upper confidence bound below target. Under the complete null every promotion is false, so this equals the false discovery rate there.
   - **Mixed signal and noise:** the false discovery proportion among promotions.
   - **Power:** the rate at which planted signals of known strength are promoted.

   Passing establishes calibration against these model families, not universal error control.

3. **Single-agent research.** One agent proposing mechanism-backed hypotheses, compared against random or genetic search through the same gate. Zero survivors is an acceptable outcome; the success criterion is calibration plus an honest comparison with the baseline. The real campaign is a pipeline validation run, since its post-cutoff holdout is too short for power.

4. **Swarm and canonicalization.** Parallel agents sharing history, e-graph canonicalization, and effective-N sensitivity analysis. Shared history adds new adaptive feedback, so a null smoke test runs at this scale; calibrated swarm error rates are not claimed.

5. **Data integration.** Add governance and social sources with immutable revisions, lineage, and per-source availability tests.

6. **Verified reference kernels.** Pre-register the theorems of Section 6, have them independently reviewed, and freeze them. Build the reference kernel, prove them, and differential-test it against the engine. Exit with the reference kernel adopted.

7. **Sources as a language feature.** Implement source declarations, binding, and certificates (Section 5), and move every existing feed onto them, deleting the per-feed code paths they replace. Exit when every existing feed is bound through a certificate, and a real new feed has been added end to end (declaration, certificate, binding, evaluation) without changing the kernel. A separate step before the next stage, under its own pre-registered criteria, then replaces the engine and the Python with an executable Lean engine whose laws are proved on the running code, with OCaml for untrusted plumbing, so that the next stages are written once, in Lean.

8. **Multi-asset data.** Build point-in-time universes and sources for equities, indices, commodities, and macro vintages; confirm which execution venues the operator may use and certify execution mappings for them; generalize the kernel from a single 24-hour perpetual to instrument kinds with their own calendars, settlement, and decision cadences; type the level and change distinction of Section 4; repeat the stage-0 feasibility check per asset class; and repeat the stage-2 calibration with null models matching the larger universe's cross-asset dependence,

judged on a locked evaluation suite of null generators kept separate from those used to design the gate. Results are reported at their claim tier, signal study or backtest.

9. **Cross-domain exploration.** Implement the exploration, confirmation, and holdout periods of Section 10. Calibrate the whole two-stage procedure in simulation, including planted cross-domain signals, before any real campaign. Zero confirmed hypotheses is an acceptable outcome.

10. **Thesis track.** Implement thesis registration, retrieval logging, the fixed sizing rule, and scoring against the baselines of Section 12. It needs only the ledger, the venue's data, and stage 0's cost model, so it starts after stage 0 and runs alongside the others. No skill claim is made before the pre-registered minimum number of resolved thesis clusters.

11. **Paper trading, then small live size.** A human decision, outside the autonomous scope. Compare live results with the gate's predictions. A large gap means the model failed somewhere: execution, costs, a regime change, parameter instability, decay of a real edge, or leakage.

Stages 0 and 2, and their repetitions in stage 8, matter most and are the easiest to skip. Before scaling the search, the system must show that its features can be reconstructed as they were available historically, and that its complete search and selection procedure meets its false-promotion targets under the specified market models. A correctly calibrated gate will still occasionally promote noise at its stated rate; the system is broken when the observed rate exceeds the target by more than sampling error. These checks cannot guarantee future profit, but they establish whether the research process meets its basic design requirements.

17 Findings to date

This section reports recorded judgments and completed implementation checks. It makes no trading claim. Its counts are observed results, not chosen thresholds. It uses no unreleased holdout, no result derived from an embargo, no observation from a protected forward window and no live thesis outcome. Stages 7–9 and C10.5 have no exit judgment to report.

Feasibility (stage 0). The replay and availability checks supported the pilot. C0.7 failed coverage in three of 44 cells: BTC sell \$1k (0.894), SUI sell \$10k (0.882) and kPEPE sell \$10k (0.817). The human authorized exit with that failure recorded, never as a pass, and stages 2, 3 and 10 inherit the cost qualification. The judged window is [2026-10-25, 2026-10-26) (`audit/stage0/criteria-result.txt`, `audit/stage0/execution.json`). An earlier, spent window had an invalid clock instrument and seven median-excess failures, which the later amendments do not erase (`audit/criteria.md`, C0.3/C0.7; `audit/decisions.md`, 2026-10-26 15:51 UTC).

Kernel MVP (stage 1). C1.1–C1.8 were met, C1.7 under its martingale amendment. The checks caught 17 mutants; the adversarial codes and hand fixtures passed; the independent timing check agreed; W0's ordered replay matched the witness. The evaluator's median time was 14.9 ms. The original canary market had arithmetic drift although its log returns had zero mean, so its failure led to a corrected market, not a looser cutoff. C1.3 compared hand values to 10^{-9}, and C1.6 used the pilot's frozen identity, which predates the certificate and extractor fields (`audit/stage1/c11-mutants.txt`, `audit/stage1/c12-adversarial.txt`, `audit/stage1/c13-fixtures.txt`, `audit/stage1/c14.txt`, `audit/stage1/c14-disagreements.md`, `audit/stage1/c15-timing.txt`,

audit/stage1/c16-ledger.txt, audit/stage1/c17-amended.txt, audit/stage1/c18.txt; final review audit/stage1/exit-review-gpt-5.md).

Holdout-test studies. Locked suite v1 failed in the sparse cells of 540 and 546 steps: rejection rates 0.0595 and 0.0602, with upper bounds 0.0648 and 0.0656 above the tolerance 0.0625. Expanded calibration v2.1 still failed on rare losses; calibrating familiar shapes did not certify arbitrary means. The human adopted the bounded-return betting test, and its fresh v3 suite passed its 99 judged pairs. That checks the implementation on bounded conditional-null series, not the bound in real markets (audit/stage2/holdout-size-locked.json, audit/stage2/holdout-size-locked-v2.json, audit/stage2/holdout-size-locked-v3.json; C2 holdout revisions).

Calibration and sensitivity (stage 2). The first mixed sweep screened nothing: the observed dispersion of Sharpe ratios across trials set the hurdle, so real edges raised their own hurdle. After a revised reference scale and an independent implementation, the r2 exit record reports C2.1–C2.5 met. Every complete-null cell had zero promotions in 300 campaigns, and every judged Sharpe mixed cell had zero FDR. C2.2's first generator comparison had failed on variance and standardization defects, which were fixed before the comparison that passed (audit/stage2/c22.json, audit/stage2/c22.v2.json, audit/stage2/c26-failure.md, audit/stage2/exit-r2.md). The second exit review retained provenance, probe-coverage and truth-verification gaps in its supplied packet (audit/stage2/exit-review-gpt-2.md); the human's exit decision is not a retrospective repair of that packet. C2.6 failed in both rounds, as written and amended: r2 N2-s3 power was zero and N2-long was 5/100; stage 2 exited only by the human's sensitivity waiver, not a pass. Its searchers omit mechanisms, so these are not stage-3 power estimates (same exit record; audit/decisions.md, 2026-10-28 13:42 UTC).

Historical-cost alternative. The archive equivalence check failed its matched-snapshot coverage requirement: about 30% of snapshots matched within one second, against the chosen 50%. It also missed its agreement thresholds: 32 of 44 live cells missed 95% cost agreement (down to 71.9%), and 4 missed 95% censoring agreement. The pre-holdout cost-table alternative lapsed. Stage 3 kept the qualified C0.7 table, which was fitted after its historical holdout (audit/stage3/archive-equivalence.json). A separate archive-regime diagnostic decided nothing, and this section does not treat it as a new judgment.

Single agent (stage 3). C3.2 v1 failed: 20 campaigns crashed before the model ran, for lack of funding inputs. V2 made no evaluations because workspace writes were denied, so it established no procedure despite its numerical count. Under repaired permissions, v3 had zero counted promotions in 20 evaluating, audited campaigns. The real two-arm run had zero finalists in both arms, so C3.3 and C3.4 were met only under the zero-finalist revisions. The release path was not exercised and the holdout remains unreleased. This is pipeline validation. It is neither evidence of an edge nor evidence of no edge, and it says nothing about agent superiority (audit/stage3/exit.md, audit/stage3/write-denied-2026-10-31.md, audit/stage3/null-summary-v1.json, audit/stage3/null-summary-v2.json, audit/stage3/null-summary.json).

Swarm smoke failures (stage 4). C4.5 v1 failed. Its first two campaigns made 33 and 36 evaluations, each below the 200 floor, and both used cross-agent history without crashing. The floor mistook agents that stopped on their own for broken machinery. V2 failed with two counted campaigns out of ten. In each, an agent ran /work/mm-agent instead of the permitted relative

command, received no evaluation and stopped. One campaign also exposed a crash in reporting a missing log. No false discovery was observed, but neither version passed. V3 is unfinished, and no verdict is reported here (`audit/stage4/null-summary-v1.json`, `audit/stage4/null-summary-v2.json`; `audit/decisions.md`, 2026-11-01 10:27 UTC and 2026-11-02 12:53 UTC).

Data integration (stage 5). Discourse and Snapshot exited as prospective-only, with unlocks blocked. The original Discourse C5.3 failed its clock and cadence requirements, and the amendment admitted no historical claimed times. Lineage and independent event-count checks passed for the retained exit data. The predecessor hash of the earlier migration was never recorded. C5.6 calibration is deferred, and it is mandatory before any search store binds events (`audit/stage5/gpt-exit-review.md`, `audit/stage5/version-identity.md`, `audit/stage5/c55-evidence.md`).

Reference-kernel comparison (stage 6). Lean was adopted. Six abstract laws were accepted, and 1,196 verdicts and 3,372 runs agreed exactly, including complete node streams. Earlier agreeing runs did not qualify: first the corpus was incomplete, then they compared digests instead of streams. L1, L3 and L4 needed recorded corrections before their proofs. Bend's concrete binding verdict succeeded but was never a reviewed frozen law. Bend had no binary64 and its one-coin smoke run exhausted memory, which prevented C6.4 differential testing and therefore eligibility. Lacking a verified integer library separately prevented proving L1 and L4. Performance favored Rust (16.0 ms evaluator versus Lean reference 632.9 ms). The recovered effort and the unenforced thread configuration remain qualifications, not repaired results (`audit/stage6/report.md`, `audit/stage6/bend.md`, `audit/stage6/gpt-exit-review.md`).

Thesis implementation, not outcomes. C10.4's independent scoring check met its revised comparison after a one-ulp disagreement in the skill ratio. A later review found that both implementations treated missing funding as zero, so the specification and both scorers changed before the next judged check. The exact-arithmetic refreeze again agreed on 1,000 sets. The independent position check left a strict-xfail defect in the managed-return summary for a thesis with one entered leg. These are implementation findings only. No live score or operational exit is inferred (C10.4 revisions; `audit/decisions.md`, 2026-10-25 01:54 UTC, 2026-11-03 00:02 UTC and 03:49 UTC).

Reproducibility and operations studies. Cross-machine test 1 differed in binary identities, not in returns. After one binary was shipped, the null, mixed, calibration and revised-gate comparisons agreed. The loader candidate reproduced the recorded searches and the reference corpus, and Indexed Jobs passed the killed-pod retry check (`audit/crossmachine/freeze.md`). The pre-freeze erasure reproduced the results, the normalized ledgers and the differential corpus (`audit/erasure/reproduce-null.json`, `audit/erasure/reproduce-mixed.json`, `audit/erasure/differential.json`). Recorder handoff retained overlapping evidence, but boundary polls exceeded the completeness limit, and its L2-based clock acceptance was later withdrawn (`audit/cluster/handoff.md`; `audit/decisions.md`, 2026-10-31 clock-acceptance correction). Shared test builds and plain worksteal retained the original cases and cut full-suite time to about 228 seconds. Disk-backed SIGKILL tests exposed a recovery defect for an empty schema, now a strict xfail awaiting refreeze (`audit/2026-11-04-suite-speed.md`).

Adversarial findings remain findings. Nine reproduced defects in the kernel, ledger and mechanism remain qualifications of the frozen pilot. They are not evidence that its proofs cover

the implementation. The defects are future-dependent delisting, reopening after stop-out, unbudgeted holds, pre-fill sizing equity, bankruptcy resurrection, boundary-fill reporting, omitted closing costs, undefined-label coverage and replacing ledger inserts. Their corrected definitions are specified but not yet judged on a replacement engine (`audit/reviews/2026-11-03-gpt-hunt-kernel.md`, `audit/reviews/2026-11-03-gpt-hunt-gate.md`; reproductions and disposition `audit/decisions.md`, 2026-11-03 04:37/15:20 UTC and 2026-11-04 00:03 UTC). The recorder, plumbing and ops hunts also found real persistence and isolation failures, with fixes recorded there. They show the need for failure-path tests. They do not establish that recording is complete or that the code is correct in general (`audit/reviews/2026-11-02-gpt-hunt-recorders.md`, `audit/reviews/2026-11-02-gpt-hunt-plumbing.md`, `audit/reviews/2026-11-03-gpt-hunt-ops.md`; `audit/decisions.md`, 2026-11-02 ops-fixes entries and 2026-11-03 03:43 UTC).

A Derivations and choices

This appendix is the home of the rationale. The specifications keep the exact executable definitions, and `audit/criteria.md` keeps the frozen criteria and their revisions. *Derived* means a calculation under named premises. It is not a universal validation. *Choice* means a policy or model convention, given with its reason. *Owed* marks a stronger claim that has not been derived. Dates, hashes, versions and measured results identify evidence, not tuned constants. The assumption inventory is `audit/reviews/2026-11-06-gpt-assumption-sweep.md`, ranked findings 1–16 and "Lower-consequence acceptance constants". Its cited gaps stay open unless a derivation below explicitly closes them. This consolidation changes no frozen threshold or judgment.

A.1 Statistics and the gate

S1. Noise ceiling (derived illustration). For independent stationary Gaussian returns near zero Sharpe, annualized $\mathrm{SE}(\hat{S}) \simeq T^{-1/2}$ over T years. Choosing $T = 5$, 1,000 null trials, threshold 1, and ten real trials of true Sharpe 0.5 illustrates selection. Here $1 - \Phi(\sqrt{5}) = 0.0127$ gives 12.7 null acceptances, versus $10[1 - \Phi(0.5\sqrt{5})] \simeq 1.3$ real ones. The share of null acceptances in the expected counts is about 90%. It is not an exact expected FDP. For independent $Z_i \sim N(0,1)$,

$$\mathbb{E} \max_i \hat{S}_i = \mathrm{SE}\, \mathbb{E} \max_i Z_i, \qquad \mathbb{E} \max_i Z_i \sim \sqrt{2 \log N}, \qquad h(N,T) = \sqrt{2 \log N / T}.$$

The leading term at $N = 1,000$ and $T = 5$ gives 1.66; integrating the maximum density $N\phi(z)\Phi(z)^{N-1}$ gives about $3.24/\sqrt{5} = 1.45$. Table 1's trial/year grids and net-Sharpe range 0.3–1.0 are illustrative choices, not calibrated scope screens. Dependence and nonstationarity make a literal independent-trial reading invalid.

S2. DSR reference and numerical rules (derived, qualified). For T per-period returns, let $\hat{S}$ be their mean divided by sample standard deviation (divisor $T-1$), $m_k = T^{-1} \sum (r - \bar{r})^k$, $\hat{\gamma}_3 = m_3 / m_2^{3/2}$, $\hat{\gamma}_4 = m_4 / m_2^2$ (ordinary kurtosis), and γ_E the Euler–Mascheroni constant. Then

$$\mathrm{DSR} = \Phi\left(\frac{(\hat{S} - S_0)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \hat{S} + (\hat{\gamma}_4 - 1)\hat{S}^2/4}} \right), \quad S_0 = \frac{(1-\gamma_E)\Phi^{-1}(1 - 1/N) + \gamma_E \Phi^{-1}(1 - 1/(Ne))}{\sqrt{T-1}}.$$

Set $S_0 = 0$ for $N = 1$. The asymptotic variance is $(1 - \gamma_3 S + (\gamma_4 - 1)S^2/4)/(T-1)$; at $S = 0$ its reference scale is $1/\sqrt{T-1}$. This requires finite variance, a martingale CLT and no dominating

observations. Even the zero-Sharpe Gaussian finite-sample variance is $(T-1)/(T(T-3))$ for $T > 3$. The numerator uses an expected-maximum approximation, not an exact maximum distribution. Costs, negative means, heavy tails, adaptivity and negative trial correlations each rule out a universal conservative reading. The observed dispersion across trials is not a null scale. Autocorrelation changes sampling behavior [23]. Annualization divides annual Sharpe by $\sqrt{252}$ for the chosen trading-day convention, $\sqrt{365}$ for daily continuous markets, or $\sqrt{6 \cdot 365}$ for the pilot.

Pearson's inequality $\gamma_4 \geq \gamma_3^2 + 1$ makes the variance term nonnegative, but a two-point boundary is numerically unstable. The relative degeneracy coefficient 10^{-12} is a *choice* to refuse scores that roundoff could change. It is not a proved error budget. Computing moments on $r/\max|r|$ preserves Sharpe, skewness and kurtosis and avoids overflow. Nonfinite returns are rejected. The revised derivation is C2.3 and the stage-2 Gate revisions of 2026-10-27; DSR ≥ 0.95 remains a chosen screen (S5), not a confidence statement.

S3. Betting, absorption and multiplicity (derived under scope). Let r'_t be integrable pre-absorption net returns with $\mathbb{E}[r'_t \mid \mathcal{F}_{t-1}] \leq 0$ while alive, and stop at bankruptcy. Choose predictable $\lambda_t \in [0, 1/2]$. With the lower bound $r'_t \geq -L$, $L = 2$, every factor $1 + \lambda_t r'_t \geq 1 - L/2 = 0$ is nonnegative and $K_t = \prod_{s \leq t}(1 + \lambda_s r'_s)$ is a nonnegative supermartingale from 1. Bankruptcy's reported -1 and the pre-absorption terminal loss cannot create a new maximum, and later reported returns are zero. Thus their running maxima agree, and Ville [32] gives $p = \min(1, 1/\max K)$ with

$$P(p_i \leq u \mid \mathcal{F}_{\text{search}}, H_{0,i}) \leq u.$$

Holm then bounds any false rejection in the frozen family at the *chosen* $\alpha = 0.05$, without assuming independent finalists. The finalists must be frozen on search information alone. The pilot's reported floor $B_{\text{rep}} = 1$ follows from absorption and zero subsequent equity (I10/L6). It does not imply this conditional null, because

$$\mathbb{E}[\max(r', -B_{\text{rep}}) \mid \mathcal{F}] = \mathbb{E}[r' \mid \mathcal{F}] + \mathbb{E}[(-B_{\text{rep}} - r')_+ \mid \mathcal{F}].$$

Conditioning on never going bankrupt does not restore it either. The corrected synthetic scope is E2. In real markets, $r' \geq -L$ and the frozen pilot's bound remain *assumed, owed*. Rare observed bankruptcies do not certify them (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 1; C4.5's 2026-11-04 qualification).

The bet is $\lambda_1 = 0$, then $\min(1/2, \max(0, S_{t-1}/Q_{t-1}))$, zero if $Q = 0$, with $S = \sum r$, $Q = \sum r^2$. Maximizing the second-order approximation $\mathbb{E}\log(1 + \lambda r) \simeq \lambda\mu - \lambda^2\mathbb{E}r^2/2$ gives $\mu/\mathbb{E}r^2$, which motivates the plug-in. The cap $1/2 = 1/L$ is derived, not chosen: Ville's nonnegativity requires $\lambda_t \leq 1/L$ under $r' \geq -L$, so any later bet must keep $\lambda_t \leq 1/L$. Only the plug-in form inside $[0, 1/2]$ is a choice. The rule as mathematics does not prove binary64 behavior for every finite input: the existing strict xfails in `gate/tests/test_gate_spec.py` show overflow and underflow of S, Q on extremes that no campaign produces. O1's provisional floor $B_{\text{rep}} = 0.2$ (a different quantity from the null's bound L) is a development candidate. It is not a safety claim or venue equivalence. It still owes complete collateral and continuation mechanics, a conditional null for the new score, and fresh calibration after comparing a better $B_{\text{rep}} = 1$ bet. Flooring saved traces is not an O1 simulation (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 2; `audit/stage2/bound-review-gpt-1.md`).

For independent campaigns each at 5%, 20 campaigns yield $1 - 0.95^{20} \simeq 64\%$ chance of any false promotion. A project-wide guarantee instead follows from a union bound when $\sum_j \alpha_j \leq \alpha$. The allocation $\alpha_j = \alpha 2^{-j}$ is an illustrative choice. Stage 3 corrects within each arm, so its two-arm union bound is twice the per-arm level, not a joint 5% claim. Missing frozen finalists stay at $p = 1$, because removing them could lower Holm's hurdles (C3.3, C9.4).

S4. Calibration sample sizes and power (derived bounds; chosen targets). For x promotions in n independent complete-null campaigns, the one-sided Clopper–Pearson upper endpoint q solves $P_q(X \le x) = 0.05$. With $x = 0$, $q = 1 - 0.05^{1/n}$: below 5% needs 59 runs, below 1% needs 299. C2.4 chooses 300 runs per cell with targets of 5% for promotion and 10% for the finalist path. Since $P_q(X \le x)$ falls in q, $q(x) \le q_0$ iff $P_{q_0}(X \le x) \le 0.05$. At $n = 300$, $P_{0.05}(X \le 8) = 0.0341 \le 0.05 < 0.0650 = P_{0.05}(X \le 9)$, so at most eight promotions meet the 5% target. Likewise $P_{0.10}(X \le 21) = 0.0458 \le 0.05 < 0.0699 = P_{0.10}(X \le 22)$, so at most 21 meet the 10% finalist-path target. At zero of 300 the bound is 0.00994. C3.2's 20 runs with at most two counted events bound only about 28% (zero would bound 14%); C4.5's ten with at most one bound about 39% (zero would bound 26%). These choices keep expensive workloads affordable. They support smoke claims only, not a 5% calibration.

For campaign FDP $X \in [0, 1]$, Hoeffding gives $\mathbb{E}X \le \bar{X} + \sqrt{\log(1/\delta)/(2n)}$ with confidence $1 - \delta$. C2.5 chooses $n = 100$, $\delta = 0.05$, target 0.20, giving $\bar{X} \le 0.20 - \sqrt{\log 20/200} \simeq 0.0776$. C9.5's union bound over M mixed cells gives penalty $\sqrt{\log(M/0.05)/(2n)}$; its minimum sample of 400 and its 0.10 target are prospective choices of resources and error policy, not a power derivation. The locked holdout suites choose 20,000 series per cell, lengths 540, 546 and 1,080, and levels 0.05, 0.01 and 0.005. Bonferroni confidence $1 - 0.05/K$ is derived to cover all K pairs (54 in v1 and v2, 99 in v3). The tolerances 0.0625, 0.015 and 0.01 are chosen slack for an implementation check. The extra 0.001 and 0.0005 rates are descriptive. These suites cannot establish conditional validity outside their nulls.

S4a. Holdout power (derived approximation, not a universal limit). At a saturated half-cap, $\log K_n \simeq n(\mu/2 - \mathbb{E}r^2/8)$, so ignoring the quadratic cost, one finalist needs $\mu \simeq 2 \log 20/n$ to reach $p = 0.05$: about 1.66%, 1.10%, 0.55% and 0.11% per step at $n = 360, 546, 1,080, 5,400$. Holm raises that hurdle. An unconstrained second-order Kelly approximation gives $\log K \simeq nS^2_{\text{step}}/2$, hence annualized $S \simeq \sqrt{2 \log 20\,(6 \cdot 365)/546} \simeq 4.9$ at 546 steps. Neither is a theorem that every valid test needs that effect. The nonexistence of a uniformly powerful test over unrestricted finite-mean distributions [2] is a different statement.

C2.6's 30% sensitivity target is a *choice*. It is not derived from operational utility (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 12). Planted Sharpes 1, 2 and 3 are chosen effect scenarios. N3's amended 2.5 reflects that 3 is unattainable on its calibration curve. The suspended 0.7%-mean cell was chosen from a grid that considered the test alone. The later N2-long cell (0.2% mean, 900 days) was chosen from a grid of reachable effects with high test power. Neither met the pipeline target, and the waiver stands. C8 and C9 still owe effect sizes, dependence, observation budgets and a sensitivity matched to their intended studies.

S5. Selection, workloads and smoke choices. DSR 0.95, SPA/StepM 0.05, stationary-bootstrap mean block 6 four-hour steps and 1,000 resamples are *choices*: a stringent screen and a reproducible, affordable heuristic for dependence. They are not cutoffs derived from a claim (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 12). The raw count of filled trials is the primary N, so that no favorable estimate is chosen. The complete-linkage $|\rho|$ cuts 0.9 and 0.5 are chosen contrasts for testing fragility, not bounds on the effective number of trials. Zero-variance trials are singletons, which avoids undefined correlations. The mechanism's positive sample-mean difference matches its stated sample claim. The 10–90% exposure shares are an underived choice to keep the claim from being vacuous. Owed is an argument about precision and power that admits genuinely rare mechanisms, not just balanced branches (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 7; C3.3).

The single-agent 100-evaluation budget, baseline 100/400, four-agent 400-total/100-each swarm, 20 history calls, up to ten agent-selected candidates, 200 turns and two hours define a *chosen* workload and comparison that the pilot can afford. They do not bound the information agents receive, and they do not guarantee adequate research opportunity (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 14). Checks are free because they run only the parser and type checker. Every eval call before the budget is exhausted counts against the 100 evaluations, including rejections, zero-fill runs and duplicates; post-exhaustion BUDGET refusals do not. The gate's N counts only distinct canonical strategies that filled at least once. History is logged because it changes later selection. The prediction is excluded from the equivalence key because identity tracks trading behavior, not an explanatory filter. Evidence from baseline searchers does not calibrate stage 3's richer set of disclosures. C4.5 v3's two evaluating agents and cross-agent receipt directly witness the swarm paths. Its tolerance for one unresolved and one unestablished campaign is a chosen smoke policy. Promotion precedence can mask a failure to establish, a gap that is reported but not repaired (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 13). C9's $1 \leq k \leq 5$ is a prospective choice to keep confirmation families small. It is not a theorem that exploration requires five slots.

A.2 Execution

E1. Grid, fills and costs (derived timing; chosen model). The pilot chooses hourly feature ticks, four-hour decisions, initial \$10,000 equity, target gross 2, \$10,000 order cap and \$1,000/\$10,000 cost buckets so that a small account is feasible on coarse history (C0.8, C1.3, I9). The 4-hour history is the finest retained execution grid. If the close at boundary B arrives at $B + \delta$, the opening price at B predates the decision, and the first recorded open after it is $B + 4$ hours. An empty selected bar does not roll forward, because rolling would silently select a different execution opportunity. Settlement precedes fills at equal instants, so that the position that actually held the hour pays for it. These conventions make no claim about queue position or executable liquidity.

For reference notional a, adverse slippage fraction s and fee f, a buy costs $as + af(1 + s) = a(s + f + sf)$; a sell's cost is $a(s + f - sf)$, bounded by the same $\kappa = s + f + sf$. The base taker fee 0.00045, hourly funding cap 0.04, size decimals, \$10 venue minimum and maintenance tiers are rules taken from the source, not fitted thresholds (C0.6; `venue/hyperliquid/rules.json`). Applying the highest documented fee and maintenance to unknown historical rules is a model choice. It is not proved conservative for a full replay. Truncation toward zero respects the venue's size granularity. Exact zero targets close dust, so no position stays below the minimum permanently. The 365-day/24-hour unit year is a fixed conversion convention, not a market calendar.

E2. Corrected gapless loss bound (derived; limited domain). Choose the account's own stop-out $\Lambda = 4$, distinct from venue liquidation. Using the same stop-out and fill-admission mechanism bounds held exposure, which merely normalizing targets does not. Let $e_{\text{prev}} > 0$ be previous recorded equity before its fills, G_0, G_1 gross at the marks before/after those fills, and m the largest next-bar carrying loss per unit mark notional through funding and adverse closeout, its cost included. Assume that opens equal marks without gaps, that any delisting closeout used in this argument is at the mark, and that $\Lambda\kappa \leq 1$. After stop-out, either nothing is held or gross at adverse prices is within Λ times equity. Price moves in the position's favor preserve that bound at the marks. A mark closeout removes notional a and costs at most κa, preserving it since $\Lambda\kappa \leq 1$. A sign-preserving reduction has the same property; other fills require margin admission, and fills only lower equity. Thus $G_0, G_1 \leq \Lambda e_{\text{prev}}$. At most one fill per coin trades at most $G_0 + G_1$, costing

$2\Lambda\kappa e_{\text{prev}}$; carrying costs at most mG_1. Therefore

$$r' \geq -\Lambda(2\kappa + m), \qquad \Lambda = 4, \quad m \leq \tfrac{1}{2} - 2\kappa \implies r' \geq -2.$$

This is reviewed algebra, not a Lean theorem (`audit/reviews/2026-11-04-gpt-nine-defects-review.md`). Real gaps break the fill-at-mark premise, and observing no excursions says nothing about the tails. The choice of 4 gives headroom above target gross 2: a single long at 2 reaches this stop-out after a one-third adverse price move, since $2(1-d)/(1-2d) = 4$. It is a policy, not venue equivalence.

Corrected synthetic markets choose $|\varepsilon| \leq 0.3$, wick multipliers $|u| \leq 0.1$ and $\kappa \leq 0.003$ as sufficient design bounds, not as the only values that work. With hourly $|f| \leq 0.04$,

$$c/o \leq (1 + 3 \cdot 0.04 + 0.3)/(1 - 0.04) < 1.48.$$

An adverse closeout loses at most $0.3 + 0.1 \cdot 1.48 < 0.448$ before closeout cost when the bar moves against the position, or $0.1 + 3 \cdot 0.04 + 0.04 \cdot 1.48 < 0.28$ when it moves for it. Closeout cost adds at most $1.1 \cdot 1.48\kappa$, so $m < 0.46 \leq 1/2 - 2\kappa$; lows remain at least $0.9 \min(o, c) > 0$. Old generators' $|\varepsilon| \leq 0.5$ also ensured positive closes, since $(1 - 0.12 - 0.5)/1.04 > 0$, but their unbounded wicks and pilot held exposure do not prove the betting bound. The clips change every null and planted path, so the old C2 calibration and betas cannot carry over (`gate/NULLS.md`, corrected-engine revision; `audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 1).

E3. Calibration choices and cost debt. C0.7's constant per coin/side/bucket is a *choice*. Sparse bars and the lack of historical full-depth features force it: one day contains only six 4-hour bars per coin, too little to condition on bar state. The one-day recording, the 70/30 chronological split, the 95th-percentile fit, the 90% coverage, at least one snapshot per minute and the 3 bps slack are *underived choices* for a tractable, conservative snapshot model. They are not bounds on costs at selected trades. For n ordered test costs, $q_{\lceil 0.9n \rceil}$ is *derived* as the least constant covering 90% (unfillable is infinite); replacing the median excess by model minus that quantile measures extra conservatism, not median accuracy. Higher costs reduce PnL on a fixed path, but equity-dependent sizing prevents extending this to monotonic full replays. For FTM and the uncalibrated thesis markets, the cell-wise maximum is an explicitly unvalidated fallback, not an economic bound (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 3). Owed are trade-conditioned coverage, a return-error tolerance, sampling uncertainty and transfer across markets. C3's doubled-cost diagnostic is a chosen stress contrast. Its relative-range coverage is only a range diagnostic, released after the primary decision. The archive matching choices (one second, 50% coverage, 95% censoring and cost agreement, 0.5 bps or 10% tolerance) were screens for proxy fidelity, not derived execution guarantees (C3 holdout revisions).

A.3 The DSL and kernel

K1. Language and numerical policies. JSON ASTs, no host code/clock/path access, strict fields, acyclic earlier-let references, explicit time-unit conversion, undefined rather than invented missing values, and evaluation of every node at every tick are *choices* that make causality, state and the information boundary checkable. Nonfinite node outputs become undefined, which prevents nonfinite targets. Undefined weights hold the position instead of inventing a signal, and the existing positions must reserve gross budget. Sizing uses post-fill equity, so costs cannot leave the next target on a budget that has vanished. Bankrupt accounts and their pending orders cannot revive within a reporting window. A boundary fill belongs to the window whose returns bear its costs,

and a closing fill belongs to the exposure it closes (a reversal belongs to the new exposure). The mechanism therefore compares fully net returns. Population window variance, EMA $\alpha = 2/(n+1)$ and first-defined-value seeding are conventional smoothing choices. They guarantee nothing about stationarity. Conversion interning includes target units: equal numeric factors do not make two units interchangeable (the t22 checker hole). Distinct level and change types are planned, and the pilot does not enforce them. The proofs quantify over numeric structures because binary64 algebra is not field algebra. The executable comparisons cover sampled binary64 behavior and are not a refinement proof.

The 1 MiB text, 4,096-node, depth-64 and 4,380-hour limits are *choices of resources and scope*, not causality requirements (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, lower-consequence DSL limits). JSON guards at bracket depth 136 and parser depth 128 are conventions for refusing input in the implementation. They guarantee no extra expressiveness. Window running sums are recomputed every window length, which trades cost against accumulated roundoff. The exact algorithm and order remain in `reference/SEMANTICS.md`. The portfolio's 10^{-12} scaling slack is likewise a chosen floating-point guard without a full roundoff budget. IEEE sums from -0.0, signed-zero tie behavior and constant normalization follow the toolchain's semantics. They are not algebraic simplifications. C1.7's Gaussian log-return mean must be $-\sigma^2/2$, *derived* from $\mathbb{E}e^X = e^{\mu+\sigma^2/2}$ so that prices are a martingale. The chosen 2% volatility, 550 days, admitted and leaking Sharpes of 4 and 20, six fixed and five fresh seeds and 20 random programs have no joint calibration of false alarms and detection (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, lower-consequence C1.7).

K2. Exact rewrites and rejected laws. R v1 is a *choice* to use only identities that are justified locally. On finite stored binary64 values, double negation restores sign bits, repeated absolute value clears them once, and absolute value after negation clears the flipped bit. Multiplication/division by dimensionless 1 and subtraction of same-unit $+0$ are exact, including -0; the divisor cannot introduce undefined. Add/multiply commute bitwise and their unit rules are symmetric. Strict and/or commute because both operands are read; a typed boolean is 0/1, so double not is exact. Swapping branches while negating the condition selects the same value (undefined condition stays undefined); both branches' state still ticks. Comparison dualities compare the same finite operands. Propagation of undefined values, and unchanged inputs to the stateful descendants that remain, lift these local identities to trading behavior. L3 proves the abstract version under these obligations. It does not prove IEEE arithmetic itself.

These counterexamples explain what is excluded and do not broaden R. Associating $(10^{16} + (-10^{16}))+1$ gives 1 versus 0; distributing $0.3(0.1+0.2)$ changes rounding by one ulp (0.09000000000000001 versus 0.09); $-0 + (+0) = +0$, so additive identity is unsafe; min/max ties on opposite signed zeros depend on operand order. An undefined c makes $\text{if}(c, x, x)$ undefined, and strict $\text{and}(\text{false}, c)$ undefined. Demeaning $[1, 2, 4]$ leaves a mean about $-1.5 \cdot 10^{-16}$, so a second demeaning changes bits. Converting 0.9452706955539223 by 8760 then back yields 0.9452706955539222. Moving lag across windows/xs changes recompute epochs or membership; $\text{neg}(\text{sign}(0)) = -0$ differs from $\text{sign}(\text{neg}(0)) = +0$. The implemented R v1 contains no lag-fusion law, even though C4's candidate list proposed one. Definitions: `kernel/CANONICAL.md`; C4.1, L3.

K3. Canonicalization, tolerances and corpus debt. Limits on nodes and iterations instead of wall time, then extraction of the fewest nodes with a lexical serialization tie-break, are *choices* for reproducible keys. They do not give complete semantic equivalence. The chosen 30 iterations/20,000 nodes and C4.2's 99% over 1,000 generated/20 hand pairs (up to ten rewrites) have no coverage-risk derivation (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, lower-consequence C4.2).

Identity pins versions and float bits because limits and rounding can change keys. Append-only transactions precede disclosure, so a failed commit cannot leave unlogged research. Result hashes detect stale corruption. They do not detect an adversary who can rewrite the schema and the hashes. The operating-system boundary must withhold writable ledger connections (I6/I7).

I5's *chosen* tolerance is $10^{-9}\max(10^4, |e|, \max_{s<t}|e_s|)$: reconciliation terms keep the roundoff of the magnitudes they once attained, even after equity collapses, so scaling by current equity rejected correct runs. The coefficient is not derived from a bound on operation count and can hide large later errors (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, lower-consequence I5). C6.4's normal comparison is bit exact; the optional differing-libm envelope $\max(10^{-12}, 10^{-9}\max(|x|, |y|))$, C1.4's \$1e-6, and C1.3's \$1e-9 are chosen comparison budgets. They do not establish mathematical equivalence. No C6.4 libm exception was used. C10.4's relative coefficient belongs in T3 below, never in a discrete decision.

C6.4's 550 days gives 13,200 hourly ticks: enough for three nested 4,380-hour lags, not four (17,520 hours). So "long enough for nested lags" is not established for the language. Fresh C7.7 and C8 corpora owe each program's end-to-end dependency warm-up and defined-state coverage. They cannot inherit that number (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 5). Sample counts in C1.1–C1.5 (one mutant/invariant, 30 attacks/15 mechanisms/ten independent attacks, eight named hand scenarios, 1,000 traces at least half violated), C4.1 (1,000 rewrites/three markets), C6.4 (1,000 generated programs), C7.4 (first 200 baseline programs), C7.6 (1,000 rule/query cases, 100 per category), and C8.1 (100 dates) are *chosen test inventories*, sized to be tractable while covering varied adversarial and boundary cases. They do not bound the prevalence of defects not yet observed. Category coverage and independent expected outcomes matter separately (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, lower-consequence C7/C8 counts). C1.8's 100 ms/20-run median and C6.5's five-run median are chosen checks of pilot throughput and stability, not correctness conditions. Lean's tie-break is an explicit preference for maintenance and tooling, not a performance conclusion (`audit/decisions.md`, 2026-10-23).

A.4 Data and sources

D1. Synthetic no-edge construction (derived; model choices explicit). Under the simulator's settlement order, a predictable position q held through a bar earns

$$q(c-o) - qo(f_1+f_2+f_3) - qcf_4.$$

Set $c = o(1+f_1+f_2+f_3+\varepsilon)/(1-f_4)$; substitution gives $qo\varepsilon$. Fresh conditionally symmetric noise, even symmetrically clipped, has $\mathbb{E}[\varepsilon \mid \mathcal{F}_o] = 0$ given the entire observable cross-coin past. Quantity, cap, rounding and corrected fill admission are measurable at that point, so conditioning gives zero expected gross PnL before absorption. Long liquidation adds $q(\text{low} - c) \le 0$, short liquidation adds $q(\text{high} - c) \le 0$, and fees/slippage only subtract; pre-absorption net conditional PnL is nonpositive. Reported absorbed PnL need not be nonpositive (S3). Event inputs bootstrapped from an independent random stream reveal no future innovation, so the argument holds. Replaying price-linked events without that independence would break it.

N1's 2% Gaussian scale/8-sigma truncation, N2's GARCH and t_5 with common factor, and N3's two-state volatility/correlation plus symmetric jumps are *model choices* that span simple, persistent, heavy-tailed and regime-dependent volatility, all without a predictable mean. Twelve coins, 100 trades/bar, lognormal volume log-mean 10/log-sd 0.5, Gaussian wick scale $\sigma/2$, volatility cap 0.0625^2, jump cap 0.2 and five-sigma jump selection are chosen finite simulation envelopes. They are not empirically sufficient market coverage. E2 gives the distinct corrected clips. Scaling truncated innovations by their actual truncated standard deviation is required for the stated

unit variance. N3's specified Laplace truncation differs from the frozen generator's clipping, as existing strict xfails record (`gate/tests/test_gate_spec.py`). Symmetry survives, but the jump distribution and moments differ. This appendix does not silently redefine the specification. Variance targeting uses $\omega = s^2(1-\alpha-\beta)$, *derived* from stationary GARCH variance; it keeps near-integrated fits from overstating that target. For unit-variance independent factors $x_i = \lambda z_m + \sqrt{1-\lambda^2} z_i$, correlation is λ^2, motivating the square root of fitted mean correlation (flooring it at zero is a nonnegative-common-factor modeling choice).

Funding and premiums are resampled jointly in stationary blocks of mean length 168 hours, which retains their observed floor, tails and cross-coin dependence. Events use 24-hour blocks, a choice of daily scale. What establishes no edge is exogeneity to fresh price noise, not these block lengths. N3's half-probability regime assignments and five-sigma jump estimators are fitted-model conventions. Histories with fewer than 1,000 traded bars get median parameters substituted, a chosen fallback for stability. The planted two of twelve coins, funding scale 10^{-4} about interest floor $1.25 \cdot 10^{-5}$, and oracle weight 0.1 are chosen to make sparse alternatives that can be discovered. For $X = \sigma x + \beta g$, clipping at $\pm a$ changes its expectation by $\mathbb{E}[(-a-X)_+] - \mathbb{E}[(X-a)_+]$; tail probability alone does not bound tail magnitude. So the former shortcut for the bias, $aP(|X| > a)$, is not used. Instead β is calibrated through the kernel on at least 200 simulated years, which targets net Sharpe, not the unclipped algebra. A 99% positive-mean interval on 200 fresh years is a chosen truth proxy for C2.5, not analytic truth. C2.1's 200-year probes, $|z| < 3.7$, regression $p > 0.001$, and C2.2's 100-year moment tolerances (5% mean absolute innovation, 20% kurtosis, 0.05 autocorrelations/correlations) are chosen stochastic diagnostics. They prove nothing for every strategy. HC3 addresses heteroskedasticity. Per-step returns avoid a statistic of compounded dollars, whose skew made a Gaussian z-test misleading (`gate/NULLS.md`; C2.1/C2.2).

D2. Pilot scope, availability and source choices. Funding dislocations are the *chosen* pilot because settled funding, premiums and bars have a public accounting link, unlike predicted funding. W0's 10^{-4} threshold, 0.1 total target and sign reversal are a fixed witness, not an optimized edge. The two-year 730-day window, the universe of the top twelve coins by volume ranked in the preceding month, and the last 180 days reserved are choices of feasibility and breadth under retention and compute constraints. Ranking in the preceding month and including delistings avoid selecting today's survivors, but they do not derive affordability. Stage 3's historical window of 60 days (360 steps) with a one-month margin after the cutoff was chosen for fast pipeline iteration, not for independence or useful power. Costs fitted in October rule out calling that history strictly untouched. The later forward reservation of 91 days (546 scheduled steps) follows the cost-table freeze and a larger buffer after the cutoff. It remains protected, and no result is inferred here (C0.9, C3 holdout revisions).

The *chosen* $\delta = 120$ s, funding rule $\max(H+\delta, \text{reportedTime}+60\text{s})$, 30-second polling, 35-second boundary/60-second L2 gaps and 24-hour window separate lateness in the sampled source from recorder opportunities that were missed. A 1 s clock budget is small relative to the nominal slack. Neither that comparison nor a day of live measurements proves a historical availability rule. The 1% missing limit, the 24-hour longest gap and 99% replay completeness are *underived choices of feasibility*, and the missing data could contain every relevant trade (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 6, 15). Owed are coverage of the witness's dependencies and the resulting bounds on return error, before C8 annexes inherit these limits. The 1,000 boundary and random store queries and the 24-hour refetch are chosen probes of the implementation and of revisions. They are not coverage theorems. Resolving the snapshot first avoids using a superseded availability rule, but it can hide earlier values when a later correction exists. Bitemporal delivery in

stage 7 is a separately frozen fix, not the pilot's semantics.

Declaration versus binding, verifier-issued certificates and transitive lineage are *choices* that prevent a feed builder from certifying itself or converting absence into zero. Positive covered-zero evidence is needed because unseen events and no events are different observations. Terms require an affirmative grant or recorded operator decision; silence/403 is not permission. Stable upstream ids and validity intervals prevent ticker collisions. Mapping precision 0.8 was chosen after measured precision but before returns; a 50-item blind sample has no confidence/dilution target. Owed are error weighted by the consuming signal, sample uncertainty and recall beyond the sampled forums (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 9). C5's 20 historical items, 72-hour recording, 30 observed items for a claimed-time rule, five-minute cadence, 60-second Discourse prospective margin and maximum-half-rate rule are *chosen* budgets for feasibility, politeness and evidence. They are not guarantees of historical availability. The C5 amendment and absent search datasets defer calibration; they do not pass it.

D3. Scope screens, licensing and remaining prerequisites. Stage 8's fifteen-year histories and 750 consecutive fixings are *underived choices of scope*, motivated by long-history search. They do not come from S1's decision rule. Owed is power for the intended frequency, the role of the feature and target, dependence and the confirmation budget. For example, 180 monthly observations need not dominate a shorter daily series (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 4). The certificate minima for new sources and classes, the execution mappings, the aggregation floors for report cells and the C8 and C9 annex parameters are prerequisites that remain blocked. They must be derived before judged use, and this appendix invents no values for them. The tiers of signal study and executable backtest are choices that stop benchmark predictability from being claimed as trades. Artificial score bounds and conditional nulls must be derived separately. They must never come from unsupported clipping or debt cancellation (C8.4/C8.5).

Kernel-only storage/deterministic processing with aggregate-only disclosures is the human's *chosen licensing reading*. It is not proof of legal correctness or of resistance to reconstruction. Frozen reports per source, calendar restrictions and attack tests address the reconstruction of rows. No universal aggregation number has been established. Underlying histories do not certify HIP-3 execution. Free/credited sources precede purchases, and no vendor-permission campaign or paid equities purchase is authorized (`audit/decisions.md`, 2026-10-26 licensing boundary, 2026-10-31 manifest limitation, 2026-11-02 purchases).

A.5 The thesis track

T1. Sizing, scope and comparator (derived formulas; chosen policy). For invalidation distance L, nominal loss $NL = b$ implies $N = b/L$, and reference-price quantity $q_j = w_j N / P_{j,e}$. A two-leg spread therefore has gross notional $2N$. The *chosen* initial equity \$10,000, per-thesis budget $b = 1\%E_r$, reservation ceiling $5\%E_r$, $L \in [1\%, 50\%]$, probability $[0.01, 0.99]$, 72–1,440-hour midnight horizons and 90 completed daily bars define a manageable multi-day paper experiment. They do not establish safety or comparator precision (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 11). The cap reserves nominal budgets at admission, including pending entries. Nothing follows from it about resizing when equity falls, a bound on realized loss, or a calculation of dependence or gaps. Five low-L spreads can have gross $10E$. Probability alone lacks payoff magnitudes and dependence, so it is scored and not used for size.

The historical comparator uses 1,095 days and at least ten nonoverlapping complete horizon windows as *chosen* conventions for recency and minimum evidence, with no derived stability target. If x of m outcomes are positive, a chosen uniform Beta$(1, 1)$ prior gives posterior mean $(x+1)/(m+2)$:

the add-one formula is derived, but its prior and eligibility floor are not calibrated. Nonoverlap reduces mechanical reuse of data, not dependence. Daily candles are a historical comparator, not a reconstruction of live builder marks. Missing windows stay missing instead of moving the anchors to improve coverage. These choices still owe a horizon-specific comparator error/coverage target (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 11; `thesis/SCORING.md`).

T2. Scores, comparisons and sample floor. For binary o, Brier $B = n^{-1}\sum(p-o)^2$ is a *chosen proper score*. If the true probability is q, the expected loss is $(p-q)^2 + q(1-q)$, which is minimized at $p = q$. Skill $1 - B/B_{\text{ref}}$ normalizes improvement to a comparator; if $B_{\text{ref}} = 0$, a ratio is undefined, not perfect skill. Pairing identical resolved tasks keeps missing comparator coverage from silently changing the evaluated population. The conditional ablation asks about reading for probability assessment of the main agent's selected task; the full no-reading arm also tests task selection. A fixed 50% comparator ignores chosen exposure, drift, carry and costs. Outcome-market matches and benchmark forecasts answer different tasks and remain descriptive.

Choosing a two-sided 5% hit-rate test and 80% power, a normal approximation equates effect $\sqrt{n}(p_1 - p_0)$ to the null critical deviation plus alternative power deviation:

$$n \simeq \left[\frac{z_{0.975}\sqrt{p_0(1-p_0)} + z_{0.80}\sqrt{p_1(1-p_1)}}{p_1 - p_0}\right]^2.$$

At the *chosen illustrative* $p_0 = 0.50$, alternatives 0.60/0.55 give about 194 and 783 independent trials. This does not derive 200 dependent graph clusters for paired Brier skill. C10's floor of 200 resolved clusters is a *chosen prohibition on early skill claims*. It does not permit a claim at 200. An inferential protocol still owes a comparator, weighting, dependence, multiplicity, stopping and power. Connected components over session, catalyst and overlapping legs deliberately merge the obvious dependencies, but they do not prove independence. The 95% percentile intervals from 10,000 resamples with PCG64 seed 10, omitted below two resolved clusters, are reproducible descriptive choices. They are not certified for coverage (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 11).

T3. Timing, exactness and operational quotas. Hourly execution rather than four-hour pilot execution is a *choice*, made because theses last several days; $\delta = 120$ s prevents a decision on a core close from using its simultaneous open. Builder marks sampled at $H + 5$ s and sealed at $H + 5$ min are chosen as immutable reference conventions, because trade candles can be absent. The chosen seal is not an executable quote or proof that the mark is fresh. Missing prices never move fills backward. At an hour, the previously held leg is funded before exits and entries, so an entrant is not charged past funding. Frozen entry quantities make the as-registered probability outcomes independent of discretionary management. Exact rational arithmetic over recorded decimal values makes the boundaries of cost cells, invalidation, admission and return sign exact, so binary rounding cannot widen thresholds. C10.4's $10^{-12}\max(1, |a|, |b|)$ comparison is a *chosen* implementation tolerance that scales with the values compared. It was motivated by a one-ulp disagreement on a large ratio and is not a complete error bound. C10.3's \$1e-9 money tolerance is likewise chosen and is never used for a binary outcome (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, lower-consequence tolerances).

Daily 00:10 sessions, 100 turns/one hour, at most two registrations per main/no-reading arm, Monday benchmark batches, five named benchmark exposures, 20-day direction, 21-day horizon and $L = 0.10$ are *chosen* workloads and contrasts that are cheap and reproducible. They are not forecasts selected for performance. C10.5's 28 days, 20 main, ten resolved and four batches, and its

two-hour resolution deadlines (90 minutes for publishing a missing status), are *underived operational quotas*. Four weeks supplies calendar opportunity, but it neither proves live path coverage nor is necessary for it. Owed is a claim about operating capability that covers successful ablations, reading, spreads, builder references and invalidation, and not activity alone. The frozen window has started and cannot be rejudged retrospectively (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 8). Public GET-only retrieval, at most five validated redirects and one request per five seconds per host are chosen limits for security and politeness. Hash-frozen versions never pool, because behavior changes what was observed, selected and scored. No live outcome is reported by this appendix.

Refreeze admission and sampling capacity. The venue's documented hard limit is 1,200 REST weight per minute per IP, but our source certificate commits *all our co-egress processes together* to at most 600/min (`venue/sources.toml`, Hyperliquid). The hard limit does not authorize spending above that commitment. The deployed recorder's configured share is 480/min, now defined once as `ingest.hl.RECORDER_WEIGHT_PER_MIN` and consumed by both the deployment command and the thesis allocation. The most the thesis alone may take alongside it is therefore $B = 600 - 480 = 120$/min, shared by history, registration, discovery and marks. Round two's 720 premise was wrong and is removed, not approved as a terms change. A second report process can construct its own track and fetch; desktop/node universe, rules and backfill jobs also share this IP. The certificate's one-off 600/min scripts require every other consumer stopped. Restoring the allocation ceiling is not proof of actual shared-IP conformance: the deployed recorder can incur post-response item debt, and separate processes have separate limiters. Their accounting, exclusivity and allocation remain refreeze blockers, not assumptions a process-local limiter can enforce.

The venue charges 20 base weight per permitted info call, plus one per 20 funding rows or 60 candles. The documented/recorded page ceilings are 500 funding rows and 5,000 candles (the certificate distinguishes documentation from observation). Only the thesis exchange reserves weight *before* each attempt, for the most rows the requested range can return: one funding row an hour or one candle an interval (the unclosed last included), $\lfloor(\text{end} - \text{start})/\text{step}\rfloor + 1$ capped at the page ceiling, so a full page reserves $20 + \lfloor 500/20\rfloor = 45$ or $20 + \lfloor 5,000/60\rfloor = 103$, a day's funding tail 21 and a candle tail 20; any row beyond the reserved count is charged on receipt, so no debt is left. Both failed and short attempts consume their reservation. The shared recorder client keeps its deployed base-before-send, actual-items-after-receipt accounting. Keeping unused thesis reservation until expiry is a conservative accounting choice. A 200-day hourly page has $200 \cdot 24 = 4,800$ candles and reserves $20 + \lfloor 4,801/60\rfloor = 100$, which fits the 120 allowance. An impossible charge is rejected immediately, never waited on.

Demand must also fit, not merely each page. The chosen 1,125-day funding scope contains $1,125 \cdot 24 = 27,000$ hourly rows: 54 full 500-row pages plus a terminal short page reserve $55 \cdot 45 = 2,475$ weight per market. At most $\lfloor 120/45\rfloor = 2$ such reservations fit in a rolling minute, so the thesis can take 90 reserved weight/min of the 120, and a cold market needs 27.5 minutes. A session that read four cold markets (two registrations of two-leg spreads) would need 220 pages, 110 batches and at least 109 minutes against its one hour; this is why a session never fetches history. The history is filled *outside* sessions, page by page, and a session asks the venue only for the rows its own account depends on.

The design (`audit/reviews/2026-11-07-gpt-funding-fill-design.md`, prewarming rather than deferred base rates) keeps every registered quantity unchanged: p_{ref} and the benchmark's base-rate forecast are still computed at registration from the cached rows, and account replay still depends on the whole account's funding, closed positions included, from the earliest registration in each market. What changes is where the rows come from. The cache records the time ranges the venue has sent

whole beside the rows; a read of $[s, t)$ asks only for the part of $[s, t)$ outside those ranges below the newest fetched time, the holes among the rows inside $[s, t)$ (the frozen hourly retry, before the first row too, now limited to the range read), and the rows after the last one. Widening a read therefore finds its uncovered prefix at once, backward extension keeps every cached row, and a session read repairs nothing outside its range. The filler commits each validated page (a list of at most 500 rows, each inside the requested range, ordered, finite) before asking for the next, so a kill, a 429, a timeout or a malformed page leaves the job pending with its progress and the next attempt resumes from the committed ranges. It takes no lock a session waits on, runs through the same limiter as every other thesis read, and waits whenever a foreground charge waits or a session or snapshot holds the limiter (weight it already spent stays spent, so a foreground read can wait out up to one minute of it). A market is *ready* when its fetched ranges cover its scope up to the last UTC midnight, judged from the ranges, never from a page's length: a short page covers its range, and the hours the venue left out inside it stay explicit holes, retried daily by the filler and hourly by session reads. Coverage to the previous midnight suffices for registration because a base-rate window ends at the latest daily open the snapshot shows, the previous midnight. The freeze refuses while any listed market is not ready.

Cost, from the saved snapshot of 2026-10-28 00:11 UTC (counts only): 550 listings, 202 delisted, 348 live, 278 of them eligible (176 core, 102 builder), 70 of those with the full 27,000-hour scope. Counting the pages each eligible market's age needs (a market of h hours needs $\lfloor h/500 \rfloor + 1$ pages, the last short or empty) gives 8,618 pages; the 70 live markets not yet eligible have under 90 days, at most 5 pages each, 350 more. The cold fill is at most $8,968 \cdot 45 = 403,560$ reserved weight. At two pages a minute that is 4,484 minutes, 75 hours at full duty, about four days once the up to four session hours a day and the hourly mark windows take their share (an estimate, not a bound). Keeping it warm costs a funding tail per live market per UTC day (at most 25 rows, reserving 21), $348 \cdot 21/1,440 \approx 5.1$/min, and a leading-omission retry for the 278 live markets younger than the scope (a range wider than a page, 45), $278 \cdot 45/1,440 \approx 8.7$/min: about 14 of the 120, or 11%, before any hole. A warm session registers four markets with at most four funding tails, $4 \cdot 21 = 84$ reserved weight, under one minute. Retrying every market's leading omission hourly, as session reads do, would reserve $348 \cdot 45/60 = 261$/min, which is why the filler retries it daily (the same retry reserves 45 because its range is wider than a page); that relaxes the hourly rule only for markets no session reads. These are reservations: the actual charge of a short page is 20 plus its rows over 20.

What remains a limitation. A market that is cold when a session registers it (listed after the last pass, or a fill still running) is not an eligibility change but an explicit readiness policy: its base rate is computed from the rows ready at registration, incomplete windows are omitted by the existing rule, and below ten complete windows p_{ref} is unavailable and a benchmark registration is skipped with that reason on the record. The filler's cold-fill duration is the time until the freeze, and the initial ∼four days are not covered by any session guarantee. The rest of the steady-state workload, derived from the same reservations at $B = 120$: hourly sampling takes $20(1 + d) \leq 120$ once an hour and drains the minute before it, about 2 of every 60 minutes; the background fill about 14/min; resolution reads a few markets every five minutes; and each registration, positions or exit read of an account holding M_c core and M_b builder markets reserves, on the first read after an hour boundary, $(20 + 21)M_c + 21M_b$ (a builder market's hourly bars are its marks), about $41 \cdot 40 = 1,640$ weight, 14 minutes of the allowance, for 40 held markets, and nothing for later reads in the same hour. These fit alongside each other (about 45 of 120 per minute on average). *They do not fit the daily snapshot.* Its newest daily tail for each of 348 live listings (final only δ after the UTC midnight, so it cannot be prefetched) and its newest hourly tail for each of 176 core markets are 524 requests of 20, 10,480 weight, 87 minutes at 120/min; the daily tails alone are $348 \cdot 20/120 = 58$ minutes. The 00:10 session therefore cannot start at 00:10, whatever the funding design: the hourly tails could

be prewarmed (about 59/min, within the allowance), the daily ones could not. This is a recorded refreeze blocker, not a pass (`audit/reviews/2026-11-08-opus-thesis-refreeze-round3.md`, finding A); the options are a later snapshot time or a snapshot that does not need every listing's newest daily bar, each a criteria change. The global-coordination blocker between processes is unchanged; an in-process limiter cannot enforce it.

For due time $D = H + 5$ s, chosen once by the scheduler before sleeping, history admission stops at $D - 60$ s. The 60 s is *derived* from rolling expiry: earlier history reservations have expired at D regardless of queue length, and a slow receipt cannot introduce later item debt. Waiting releases the limiter lock; its 0.25 s reservation recheck and 0.001 s minimum expiry sleep are chosen anti-spin/responsiveness conventions, not scheduling bounds. Discovery starts at $D - 60$ s with one attempt; the last successful inventory is persisted for outages. Discovery plus d initial dex calls needs $20(1 + d) \leq B$, so at $B = 120$ the initial wave fits only $d \leq 5$. The benchmark names two builder dexes, requiring 60, but this is not evidence of today's complete inventory; that count was not read from live recordings or certified in this build. Runtime discovery records it. Above five, an alarm explicitly records that the every-dex-at-D criterion cannot hold; quota-sized later cohorts still save what fits, rather than aborting all sampling or favoring a benchmark by special case. This criterion failure remains a blocker, not a pass. Even within capacity, discovery must finish before D, the process must be scheduled and the inventory already discoverable; a cold discovery outage or a newly appearing dex cannot be guaranteed away.

Requests run in concurrent quota-sized cohorts, in lexical dex order, with unattempted dexes ahead of failed ones. At most six cohorts and exponential sleeps $1, 2, 4, 8, 16$ s (sum 31 s) are chosen transient-error policy, not evidence of availability. After each sleep, admission refuses another cohort if the 60 s rolling quota wait plus a chosen 10 s phase buffer would reach the $H + 300$ s seal. The 60 s follows from admitting no more than one minute's quota per cohort; the 10 s is the client's per-phase timeout, not a whole-request duration bound. HTTPX can spend that timeout separately on pool, connect, write and read waits, and a streaming receipt can take longer still. Thus this is a conservative start policy, not a guarantee of receipt by the seal. Late receipts stay archived and never become candidates. Every exit releases the history reservation; if the batch never starts, it ends at the seal. Every sent attempt, including malformed 200s and transport failures, remains archived. Only structurally complete HTTP-200 snapshots are seal candidates; an invalid individual mark in the selected valid snapshot remains missing without substitution. Reads incrementally index complete archive lines once across markets rather than repeatedly parsing raw payloads. None of these development fixes rejudges the running frozen track (`audit/reviews/2026-11-07-opus-thesis-refreeze.md`).

A.6 Operations

O1. Recorder capacity and timing debt. Half the documented 1,200 request-weight/minute limit gives nominal 600; reserving 120 for L2 leaves 480 for boundaries. Eleven live coins each need funding and bar requests of weight 20: $22 \cdot 20 = 440$ before retries. A deployed total 480 with the same L2 reservation instead leaves 360, an 80 deficit, requiring at least 560 total even before retries/overhead. This *derived mismatch* disproves the stated completeness capacity, and not merely a guessed threshold (`audit/decisions.md`, 2026-11-02 08:09 UTC; `audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 15). The merge by 30-second poll slot, local host first, is a chosen deterministic redundancy policy. Choosing a boundary host by completeness, not by earliest value, avoids selecting on measured lateness.

A 10-second HTTP timeout is a *choice*, but its former rationale counted a single timeout and omitted the six attempts: $6 \cdot 10 + (1 + 2 + 4 + 8 + 16) = 91$ seconds before limiter waits, already

beyond the completeness gap. Three failed verification attempts cap how long the queue blocks, not whether it can recover. Clock logging every five minutes, ten-minute bracketing gaps and forty-minute numerical changes were chosen as instruments of freshness. Repeated identical valid offsets need not mean stale measurements. Maxerror is a bound derived by the daemon, not a measured offset or freshness. These figures still owe the latency of the whole retry path and valid arguments for clock freshness (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 15).

The 240-second source target leaves slack for jitter under a five-minute coverage rule. But the capacity of the latest page (twenty Snapshot proposals), fifteen-minute GDELT polls and three-hour sampling of alerts and outcomes do not prove survival of bursts or transient failures. Owed are bounds on pagination and bursts and positive coverage specific to each source, not HTTP success (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 10). Recording before the annex, with views for integrity only, is a choice. It preserves evidence that cannot be refetched and keeps it from selecting research. It does not grant historical availability.

O2. Scheduling, preservation and budgets (choices; not guarantees). Pacing at 0.8 utilization reserves 0.2 for the human. The largest previously measured campaign usage predicts the need, with an unmeasured fallback of 0.5. These are *choices*. They are not upper bounds on the next campaign or guarantees against exhaustion (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 14). Pool sizes use measured peak memory and free capacity. Shipped content-addressed binaries and inputs keep machine paths and mutable tags from changing experiment identities. When infrastructure kills a cluster index, the same deterministic campaign reruns after partial output is handled. A campaign's own crash still counts against calibration. Neither resubmission nor a new name can restore spent evidence.

Pull-only recording on an independent host, append-only prefixes, atomic file publication and SQLite snapshot checks are *preservation choices*. They retain prior bytes and separate a live writer from a verifier. The mirror's twenty-minute allowance is derived only as ten minutes to the next scheduled run plus a *chosen, underived* ten-minute copying allowance. Owed is a bound on throughput, backlog and recovery point. A valid old mirror is not proof of freshness. Equal-size/equal-mtime corruption is outside the Oracle pull's quick-check guarantee (`audit/decisions.md`, 2026-11-02 08:32 UTC).

The replacement capacity guard (`ops/diskguard.py`; deployment status in `CLUSTER.md`, section 5) uses one rule everywhere: alarm when writer-available bytes *or inodes* are at or below 0.2 of the filesystem's total. One fifth is an explicit *precautionary choice*: leave a substantial minority for intervention, accepting earlier alarms rather than waiting for near exhaustion. It is not a growth-derived response guarantee. A fractional reserve scales with the real filesystem, unlike a fixed byte allowance, and includes inode exhaustion and blocks unavailable to the writer. At the byte threshold, constant growth g on capacity C would leave $0.2C/g$ time; adequate warning would require this to exceed detection delay plus human response time. Neither per-filesystem growth nor the human's response time has been measured, so that inequality is not established and bursts can exhaust the reserve (`audit/reviews/2026-11-06-gpt-assumption-sweep.md`, 16). No recording is deleted and no copy is stopped by the alarm.

The existing hourly user timer measures this PC's `data/` and mirror destination directly and reads each of the five prodesk7 volumes' available byte/inode counters from its atomic mirror manifest. Missing, invalid, future or more-than-30-minute-old reports also alarm, preventing a failed, killed or blocked copy from leaving an old healthy capacity report trusted forever. Thirty minutes is a *chosen* tolerance of three ten-minute mirror periods, not a throughput bound. With an hourly checker, nominal stale-alarm delay while logged in is at most $30 + 60 = 90$ minutes from the

last report, *plus scheduling and check delays*; it is not a hard bound or proof of a recovery point. Each failure emits a recurring critical, non-expiring desktop notification and nonzero exit, replacing the status-file-only alarm. Delivery requires a functioning desktop notification service; failure to deliver fails the user service. While the PC is off or the user logged out there is no alarm; the next check rejects stale reports. Deployment must confirm that the human sees a synthetic alarm. Until measured growth and an agreed response budget exist, or an independent always-on notification channel is installed, these limitations prevent a recording-safety guarantee.

Dollar limits are *operator budget choices*, not empirical thresholds: Oracle $1 guard; Hyperliquid archive $20 monthly/total authorization; Databento $100 cap within the $125 sign-up credit, no card charge; paid equity/index sources indefinitely blocked (`audit/decisions.md`, 2026-10-30 transcribed decisions and 2026-11-02 purchases). Resource caps and a 12 KiB status cap keep workloads and status bounded. They do not make anything scientifically valid. Worksteal is preferred within the human's chosen 10% margin in which the simpler option wins, and it compresses the scheduling machinery. The test-speed study supplies its measurement and makes no universal claim about speed (`audit/2026-11-04-suite-speed.md`). Frozen SCORING/DSL and off-branch stage-7 specifications retain duplicated rationale until their authorized refreeze or landing. This paper is the rationale's home now. It does not authorize forking any of them.

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